

Advanced Fluid Neutral boundary conditions: theory and implementation in SOLPS-ITER

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List of Symbols

Constants

e Elementary charge, $1.60 \cdot 10^{-19}$ C

Geometry/domain

$\mathbf{r}$ Position vector

$\mathcal{V}$ Volume

$\boldsymbol{\nu}$ Inward pointing surface normal

∇ Spatial gradient

Magnetic field

B Magnetic field strength

b_θ Poloidal magnetic field pitch (B_θ/B)

b_ϕ Toroidal magnetic field pitch (B_ϕ/B)

Kinetic modeling

E_c Center-of-mass kinetic energy

f Position-velocity phase-space distribution

$\tilde{f}$ Normalized distribution

K Rate coefficient

L Macroscopic length scale

m Particle mass

$\tilde{M}$ Drifting Maxwellian

R Reaction rate

$\mathbf{v}$ Particle velocity vector

v Particle speed (L2-norm of vector)

ϵ Knudsen number

λ Mean free path

σ Microscopic cross-section

Σ Macroscopic cross-section

Fluid modeling

c_s Isothermal sound speed

E Energy density

n Number density

p Pressure

Q Energy flux density

T Temperature (Joules)

u Bulk velocity component

$\mathbf{V}$ Bulk velocity vector

V Bulk velocity speed (L2-norm of the vector)

$\boldsymbol{\Gamma}$ Particle flux density vector

Γ L2-norm of the particle flux density vector

Boundary conditions

E'	Energy of incident particle
E	Energy of recycled/reflected particle
R	Recycling/reflection kernel or probability
R_N	Fast particle reflection coefficient
$R_{m, }$	Parallel momentum reflection coefficient
$R_{m,\perp}$	Perpendicular momentum reflection coefficient
R_{n_n}	Density reflection coefficient
R_p	Pressure reflection coefficient
$R_{u_n\phi}$	Toroidal velocity reflection coefficient
R_E	Energy reflection coefficient
v'	Speed of the incident particle
v	Speed of the reflected particle
α_ν	Angle between the surface normal and poloidal direction
δ_{sh}^{pot}	Sheath transmission coefficient
$\delta\varphi$	Azimuthal angle deviation of the recycled/reflected particle
ϑ'	Polar angle of the incident particle
ϑ	Polar angle of the recycled/reflected particle
φ'	Azimuthal angle of the incident particle
φ	Azimuthal angle of the recycled/reflected particle
ϕ_{sh}	Sheath potential difference
Ω	Solid angle

Subscripts/superscripts

a	Species index
b	Boundary
cx	Charge exchange
e	Electrons
f	Fluid
F	Fast
i	Ions (state variables) or ionization (rate coefficients) and cross-sections)
k	Kinetic
m	Momentum
n	Neutral atoms
r	Radial direction
r	Recombination
se	Sheath entrance
T	Thermal
v	Volumetric
θ	Poloidal direction
ν	Normal direction
$\nu-$	Incident part (opposite direction of the inward pointing surface normal)
$\nu+$	Outgoing part (direction of the inward pointing surface normal)
τ	Tangential direction perpendicular to the normal and toroidal direction
ϕ	Toroidal direction
$ $	Parallel direction
$\perp$	Diamagnetic direction

Chapter 1

Introduction

The aim of the Advanced Fluid Neutral (AFN) boundary conditions (BCs) is to provide macroscopic boundary conditions for fluid neutral models in plasma boundary simulations, based on the same reflection and recycling databases that are used by kinetic neutral solvers such as EIRENE. This is in contrast to earlier fluid neutral boundary conditions, which usually employed simple expressions, such as zero parallel velocity, or fixed scalar recycling factors for momentum and energy.

The AFN BCs were primarily developed in the PhD thesis of N. Horsten [11] in an in-house code. Subsequently, the models have been integrated and tested in SOLP-ITER by M. Blommaert et al. [2] (structured version) and W. Van Uytven et al. [25] (unstructured version).

The first goal of this document is to centralize all information on the AFN boundary conditions, using unified symbol conventions, as the complete description of the model is now spread out over multiple references [12, 11, 24, 26, 2, 13]. The second goal is to provide users with instructions on how to use the AFN boundary conditions, how to create new AFN databases for different wall materials, and to explain detailed options for advanced users. The third goal is to clarify the link between the equations and code implementation, making life easier for researchers that wish to improve the models in the future.

The models have been tested thoroughly for deuterium (D) on carbon (C), tungsten (W) and beryllium (Be). The code can currently handle any combination of H, D or T on any wall material, but these other combinations have not yet been benchmarked as thoroughly.

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Chapter 2

How to cite the AFN BCs

This is a technical non-peer-reviewed document and should not be cited directly. To cite the AFN BCs:

- When implementing the AFN BCs in another code: cite at least [12].
- When using the AFN BCs in SOLPS-ITER code: cite at least [12] and [26].

Additional papers can be cited if desired. The list below summarizes AFN contributions with a short description.

- Horsten, N., Dekeyser, W., Samaey, G., Börner, P., & Baelmans, M. (2016). Fluid neutral model for use in hybrid neutral simulations of a detached case. *Contributions to Plasma Physics*, 56(6-8), 610-615.

AFN in 1D in MATLAB

- Horsten, N., Samaey, G., & Baelmans, M. (2017). Development and assessment of 2D fluid neutral models that include atomic databases and a microscopic reflection model. *Nuclear Fusion*, 57(11), 116043.

AFN in 2D in MATLAB

- Blommaert, M., Dekeyser, W., Horsten, N., Börner, P., & Baelmans, M. (2018). Implementation of a consistent fluid-neutral model in SOLPS-ITER and benchmark with EIRENE. *Contributions to Plasma Physics*, 58(6-8), 718-724.

Implementation of the AFN model in the structured version of SOLPS-ITER and first benchmarks with EIRENE on slab cases. Consistency with EIRENE pumping mechanism. First use of the Maxwellian approach for the radial boundaries.

- Van Uytven, W., Blommaert, M., Dekeyser, W., Horsten, N., & Baelmans, M. (2020). Implementation of a separate fluid-neutral energy equation in SOLPS-ITER and its impact on the validity range of advanced fluid-neutral models. *Contributions to Plasma Physics*, 60(5-6), e201900147.

Introduction of separate neutral energy equation for the AFN model inside SOLPS-ITER.

- Van Uytven, W., Dekeyser, W., Blommaert, M., Carli, S., & Baelmans, M. (2022). Assessment of advanced fluid neutral models for the neutral atoms in the plasma edge and application in ITER geometry. *Nuclear Fusion*, 62(8), 086023.

AFN models in unstructured version of SOLPS-ITER, density scan of slab cases and ITER cases, introduction of flux limiters and automated selection between diffusion and Maxwellian approach.

- Van Uytven, W., Dekeyser, W., Blommaert, M., Horsten, N., & Baelmans, M. (2022). Effect of drifts and currents on the validity of a fluid model for the atoms in the plasma edge. Nuclear Materials and Energy, 33, 101255.

First benchmark between AFN and EIRENE in the presence of drifts.

- Horsten, N. (2019). Fluid and hybrid fluid-kinetic models for the neutral particles in the plasma edge of nuclear fusion devices.

PhD manuscript N. Horsten

- Van Uytven, W. (2023). Development of Fluid and Hybrid Fluid-Kinetic Atom Models for Plasma Boundary Simulations and Assessment in Reactor-Relevant Conditions

PhD manuscript W. Van Uytven

Chapter 3

Theory

This chapter combines Chapter 4 of Ref. [11], Chapter 2 of Ref. [24], and additions from other papers [12, 26, 2, 13, 1] with uniform notation.

3.1 Kinetic boundary condition

Particles incident on the solid material (e.g., the target plates) are reflected with a certain probability and the velocity distribution of the reflected particles is a function of the velocity of the incoming particle, i.e., its energy, its angle with respect to the surface normal and the azimuthal angle around the surface normal [10]. Two kinds of neutrals originate at solid boundaries: neutrals that collide with the solid wall and are reflected, and ions that collide with the solid wall, pick up an electron and are recycled as neutrals. The neutrals are emitted as fast atoms or thermally released molecules. We call the recombined ions the recycled neutrals, while we call the other emitted particles the reflected neutrals. The probabilities of fast and thermal release and the distribution (energy and direction) of the recycled/reflected particles depend on the used solid material.

In general, the boundary condition for the kinetic equation can be written as

$$\underbrace{f_n(\mathbf{r}_b, \mathbf{v})(\mathbf{v} \cdot \boldsymbol{\nu}(\mathbf{r}_b))}_{\Gamma_{\nu+}^n(\mathbf{r}_b, \mathbf{v})} = \int_{\mathbf{v}' \cdot \boldsymbol{\nu}(\mathbf{r}_b) < 0} R(\mathbf{r}_b; \mathbf{v}' \rightarrow \mathbf{v})(R^i(\mathbf{r}_b) \dots \dots \underbrace{(-f_{i,b}(\mathbf{r}_b, \mathbf{v}')(\mathbf{v}' \cdot \boldsymbol{\nu}(\mathbf{r}_b)))}_{\Gamma_{\nu-}^i(\mathbf{r}_b, \mathbf{v}')} + R^n(\mathbf{r}_b) \underbrace{(-f_n(\mathbf{r}_b, \mathbf{v}')(\mathbf{v}' \cdot \boldsymbol{\nu}(\mathbf{r}_b)))}_{\Gamma_{\nu-}^n(\mathbf{r}_b, \mathbf{v}')} \underbrace{)}_{\Gamma_{\nu-}^n(\mathbf{r}_b, \mathbf{v}')} d\mathbf{v}',$$

for $\mathbf{r}_b \in \partial D, \mathbf{v} \cdot \boldsymbol{\nu}(\mathbf{r}_b) \geq 0, \quad (3.1)$

with $\boldsymbol{\nu}(\mathbf{r}_b)$ the surface normal unit vector at boundary position $\mathbf{r}_b$ pointing towards the plasma, $\int_{\mathbf{v}' \cdot \boldsymbol{\nu}(\mathbf{r}_b) < 0} \dots d\mathbf{v}'$ the integral over the velocity space of the incident particles and ∂D the boundary of the domain. The distribution of the ions at the boundary is given by $f_{i,b}(\mathbf{r}_b, \mathbf{v})$. An expression for $f_{i,b}(\mathbf{r}_b, \mathbf{v})$ is derived in Section 3.1.1. The boundary physics, i.e., the probability of reflection as an atom and the velocity distribution of the reflected particles as a function of the incident velocity $\mathbf{v}'$ is captured in the reflection kernel $R(\mathbf{r}_b; \mathbf{v}' \rightarrow \mathbf{v})$. The reflection probabilities $R^i(\mathbf{r}_b)$ and $R^n(\mathbf{r}_b)$ for respectively the ions and neutral atoms take into account the absorption of a fraction of the particles at a boundary. In Eq. (3.1), we have also defined the velocity-dependent particle flux densities $\Gamma_d^a(\mathbf{r}_b, \mathbf{v})$ for species a for the incident ($d = \nu -$) and recycled/reflected particles ($d = \nu +$).

The EIRENE code makes use of the TRIM code reflection database [17]. This gives for a certain combination of projectile (in our case hydrogen, deuterium or tritium) and solid material, the probability of fast reflection as an atom and the distribution of the reflected energy and angles with respect to the surface

normal as a function of the incident energy and angle. The remaining particles are thermally released as molecules (in our case H_2 , D_2 or T_2). However, because we do not have a fluid model for the molecules, we assume that they dissociate immediately at the boundary surface by means of Franck-Condon dissociation. Then, the resulting atoms are emitted isotropically (Lambert's cosine law) with an energy of 2-3 eV [7].

In Refs. [12] and [11] it was assumed that $R^i(\mathbf{r}_b)$ and $R^n(\mathbf{r}_b)$ are user-defined constants, and equal for fast and thermal particles. Present-day EIRENE assumes however that the thermally emitted particles are pumped first, and fast particles are only removed if the pumped fraction is larger than the thermal fraction. It will be explained in Section 3.2.5 how $R^i(\mathbf{r}_b)$ and $R^n(\mathbf{r}_b)$ must be adapted to be compatible with the EIRENE pumping mechanism. For now, we keep the notation identical to Ref. [11].

To apply the TRIM database, we have to transform Eq. (3.1) to spherical coordinates. The spherical coordinate system is defined in Fig. 3.1, where ν is the surface normal direction pointing towards the plasma, ϕ the toroidal direction (tangential to the boundary due to toroidal symmetry), and τ the other tangential direction perpendicular to the normal and toroidal direction. The speed (magnitude of the velocity), polar and azimuthal angle of the incident and reflected particle are respectively $v', v \in [0, \infty)$, $\vartheta', \vartheta \in [0, \pi/2]$ and $\varphi', \varphi \in [0, 2\pi]$, with a prime for the incident particle.

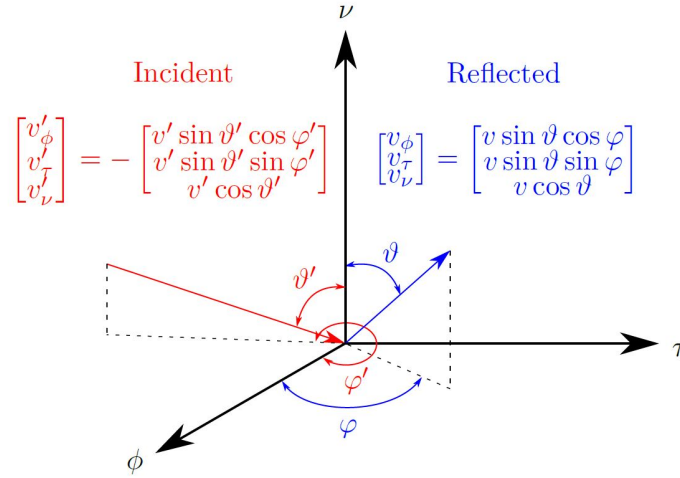


Figure 3.1: Transformation from the spherical (v, ϑ, φ) coordinate system to the Cartesian (ϕ, τ, ν) axis system.

We transform the velocity-dependent particle flux densities $\Gamma_{\nu-}^a(\mathbf{r}_b, \mathbf{v}')$ and $\Gamma_{\nu+}^a(\mathbf{r}_b, \mathbf{v})$ to their speed- and angular-dependent equivalents $\Gamma_{\nu-}^a(\mathbf{r}_b, v', \vartheta', \varphi')$ and $\Gamma_{\nu+}^a(\mathbf{r}_b, v, \vartheta, \varphi)$. $\Gamma_{\nu-}^a(\mathbf{r}_b, v', \vartheta', \varphi') dv' d\vartheta' d\varphi'$ and $\Gamma_{\nu+}^a(\mathbf{r}_b, v, \vartheta, \varphi) dv d\vartheta d\varphi$ can be considered as respectively the incident and reflected particle flux densities of species a at position $\mathbf{r}_b$ for particles with a speed between $v^{(')}$ and $v^{(')} + dv^{(')}$, polar angle between $\vartheta^{(')}$ and $\vartheta^{(')} + d\vartheta^{(')}$ and azimuthal angle between $\varphi^{(')}$ and $\varphi^{(')} + d\varphi^{(')}$. The parentheses in the superscripts indicate that the prime is optional and only for the incident contribution. The speed- and angular-dependent particle flux density follows from the relation

$$\begin{aligned} \Gamma_{\nu\pm}^a(\mathbf{r}_b, v^{(')}, \vartheta^{(')}, \varphi^{(')}) dv^{(')} d\vartheta^{(')} d\varphi^{(')} &= \Gamma_{\nu\pm}^a(\mathbf{r}_b, \mathbf{v}^{(')}) d\mathbf{v}^{(')} \\ &= \Gamma_{\nu\pm}^a(\mathbf{r}_b, \mathbf{v}^{(')}) (v^{(')})^2 \sin \vartheta^{(')} \dots \\ &\dots dv^{(')} d\vartheta^{(')} d\varphi^{(')}. \end{aligned} \quad (3.2)$$

Splitting Eq. (3.1) explicitly in the fast and thermal contributions and using the speed- and angular-dependent distributions, gives

$$\begin{aligned}
\Gamma_{\nu+}^n(\mathbf{r}_b, v, \vartheta, \varphi) = & \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_F(\mathbf{r}_b; v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi) \sin \vartheta \dots \\
& \dots (R^i(\mathbf{r}_b) \Gamma_{\nu-}^i(\mathbf{r}_b, v', \vartheta', \varphi') + R^n(\mathbf{r}_b) \Gamma_{\nu-}^n(\mathbf{r}_b, v', \vartheta', \varphi')) dv' d\vartheta' d\varphi' \\
& + \frac{1}{\pi} \delta(v - v_T) \cos \vartheta \sin \vartheta \Gamma_T^n(\mathbf{r}_b), \quad \text{for } \mathbf{r}_b \in \partial D, \quad (3.3)
\end{aligned}$$

with $R_F(\mathbf{r}_b; v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi)$ the fast reflection kernel from the TRIM database, $\delta(v)$ the Dirac delta function, v_T the speed corresponding to the 2-3 eV Franck-Condon energy (E_{FC}) and $\Gamma_T^n(\mathbf{r}_b)$ the thermal particle flux density, given by

$$\begin{aligned}
\Gamma_T^n(\mathbf{r}_b) = & \int_{v=0}^{\infty} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} (1 - R_F(\mathbf{r}_b; v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi)) \dots \\
& \dots \sin \vartheta (R^i(\mathbf{r}_b) \Gamma_{\nu-}^i(\mathbf{r}_b, v', \vartheta', \varphi') + R^n(\mathbf{r}_b) \Gamma_{\nu-}^n(\mathbf{r}_b, v', \vartheta', \varphi')) \dots \\
& \dots dv d\vartheta d\varphi dv' d\vartheta' d\varphi'. \quad (3.4)
\end{aligned}$$

In Section 3.1.1, we determine $\Gamma_{\nu-}^i(\mathbf{r}_b, v', \vartheta', \varphi')$, which is the distribution of the incident ions. This distribution follows from the plasma solution. Then, we discuss how the fast reflection kernel $R_F(\mathbf{r}_b; v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi)$ is tabulated in the TRIM database in Section 3.1.2. Finally, we explain in Section 3.1.3 how these boundary conditions are incorporated in the Monte Carlo procedure.

3.1.1 Velocity distribution of the incident ions

Before proceeding to the elaboration of the velocity distribution, we briefly introduce the reader to the sheath boundary physics. At a solid boundary, there is a plasma-wall transition layer. If the magnetic field is oblique to the boundary surface, but with an angle between the magnetic field and surface tangential larger than approximately 5° , the sheath region can be roughly subdivided in the quasi-neutral magnetic pre-sheath (Chodura sheath), with a width of the order of the ion gyroradius, and electrostatic sheath (Debye sheath), with a width of the order of the Debye length λ_D [3, 21]. The Debye length is given by

$$\lambda_D = \left(\frac{\epsilon_0 T_e}{n_e e^2} \right)^{1/2}, \quad (3.5)$$

with ϵ_0 the permittivity of vacuum and e the elementary charge. At the pre-magnetic sheath entrance, the Chodura condition states that the ion velocity parallel to the magnetic field reaches at least the isothermal speed of sound $c_s = \sqrt{(T_i + T_e)/m}$. The electric field in the magnetic pre-sheath further accelerates the ions such that also the normal velocity reaches the isothermal speed of sound at the interface between the Chodura and Debye sheath, which is called the Bohm criterion. An extension of the description of the sheath layer for very small angles between the magnetic field and surface tangential (smaller than 5°) can be found in Refs. [21, 23, 22].

The plasma fluid equations are not valid in the sheath region (Chodura and Debye sheath). Therefore, one typically solves the plasma fluid equations up to the sheath entrance. However, for the neutral simulation, we need an expression for the ion distribution at the boundary surface to evaluate Eqs. (3.1), (3.3)-(3.4). In recent years, there were many efforts in the field of modeling the sheath region [4, 5, 22]. However, we use a simplified model that neglects the deflection in the magnetic pre-sheath, similar as in the EIRENE code. It should be noted that the boundary conditions described below can be easily extended to a more elaborated sheath model, once this is implemented and tested in EIRENE.

From now on, we omit the explicit spatial dependence of $\mathbf{r}_b$ in the notation. At the sheath entrance, the ion distribution is assumed to be a truncated Maxwellian, leading to

$$\Gamma_{\nu-,se}^i(\mathbf{v}_{se}) = \begin{cases} -C^i \widetilde{M}(\mathbf{v}_{se}; \mathbf{V}_i, T_i) (\mathbf{v} \cdot \boldsymbol{\nu}) & \text{if } \mathbf{v}_{se} \cdot \boldsymbol{\nu} \leq 0, \\ 0 & \text{else.} \end{cases} \quad (3.6)$$

The constant scaling value C^i follows from the incident ion flux density:

$$\int_{\mathbf{v}_{se}} \Gamma_{\nu-,se}^i(\mathbf{v}_{se}) d\mathbf{v} = -n_i(\mathbf{V}_i \cdot \boldsymbol{\nu}), \quad (3.7)$$

in order to be consistent with the ion continuity equation. In the sheath region, the ions are accelerated perpendicular to the surface by the sheath potential difference $\Delta\phi_{sh}$. We use the following simplified model for the sheath potential difference [20]:

$$\Delta\phi_{sh} = \frac{\delta_{sh}^{\text{pot}} T_e}{e}, \quad (3.8)$$

with δ_{sh}^{pot} the sheath coefficient. If we express the particle velocity in the (ϕ, τ, ν) coordinate system of Fig. 3.1, then the velocity-dependent particle flux densities at the sheath entrance and boundary are related via

$$\Gamma_{\nu-,se}^i(v_{\phi,se}, v_{\tau,se}, v_{\nu,se}) dv_{\phi,se} dv_{\tau,se} dv_{\nu,se} = \Gamma_{\nu-}^i(v'_{\phi}, v'_{\tau}, v'_{\nu}) dv'_{\phi} dv'_{\tau} dv'_{\nu}. \quad (3.9)$$

The tangential particle velocity components remain unchanged, i.e., $v'_{\phi} = v_{\phi,se}$ and $v'_{\tau} = v_{\tau,se}$, whereas the particle is accelerated by the sheath potential perpendicular to the surface, such that $\frac{m}{2} (v'_{\nu})^2 = \frac{m}{2} v_{\nu,se}^2 + \delta_{sh}^{\text{pot}} T_e$. Elaborating Eq. (3.9), gives

$$\Gamma_{\nu-}^i(v'_{\phi}, v'_{\tau}, v'_{\nu}) = \begin{cases} -C^i \widetilde{M}(v'_{\phi}, v'_{\tau}, \dots \\ \dots - \sqrt{(v'_{\nu})^2 - \frac{2}{m} \delta_{sh}^{\text{pot}} T_e}; \mathbf{V}_i, T_i) v'_{\nu} & \text{if } v'_{\nu} \leq -v_{\min}, \\ 0 & \text{else,} \end{cases} \quad (3.10)$$

with $v_{\min} = \sqrt{\frac{2}{m} \delta_{sh}^{\text{pot}} T_e}$ the minimum occurring velocity component perpendicular to the boundary after the sheath acceleration, and $\widetilde{M}(v_{\phi}, v_{\tau}, v_{\nu}; \mathbf{V}, T) = \widetilde{M}(\mathbf{v}; \mathbf{V}, T)$.

This velocity-dependent flux density can be used to evaluate Eq. (3.1). However, the TRIM database is given in spherical coordinates (see Section 3.1.2). Hence, we need an expression of the ion speed- and angular dependent particle flux density to evaluate Eqs. (3.3)-(3.4). Using relation (3.2), gives

$$\Gamma_{\nu-}^i(v', \vartheta', \varphi') = \begin{cases} C^i \widetilde{f}_i(v', \vartheta', \varphi'; \delta_{sh}^{\text{pot}}) v' \sin \vartheta' \cos \varphi' & \text{if } v' \geq v_{\min}, \\ & \vartheta' \leq \vartheta_{\max}, \\ 0 & \text{else,} \end{cases} \quad (3.11)$$

with $\vartheta_{\max} = \arccos(v_{\min}/v')$ and

$$\begin{aligned} \widetilde{f}_i(v', \vartheta', \varphi'; \delta_{sh}^{\text{pot}}) = (v')^2 \widetilde{M} \Big(& -v' \sin \vartheta' \cos \varphi', -v' \sin \vartheta' \sin \varphi', \dots \\ & \dots - \sqrt{(v')^2 \cos^2 \vartheta' - \frac{2}{m} \delta_{sh}^{\text{pot}} T_e}; \mathbf{V}_i, T_i \Big). \end{aligned} \quad (3.12)$$

Now, we have an expression for the speed- and angular dependent particle flux density for the ions that can be plugged into Eqs. (3.3)-(3.4).

3.1.2 Reflection kernel from the TRIM database

In this section, we explain how to extract the reflection kernel $R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi)$ from the TRIM data tables. We omit the spatial dependence. Moreover, the reflection kernel becomes only spatially dependent if the vessel consists of different materials.

An expression for $R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi)$ cannot be directly constructed from the database, because the tables are given as a function of the incident energy and polar angle. Hence, we first find an expression for $R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta\varphi)$, with $E' \in [0, \infty]$, $\vartheta' \in [0, \pi/2]$ respectively the kinetic energy and angle with respect to the surface normal of the incident particle, and $E \in [0, E']$, $\vartheta \in [0, \pi/2]$ and $\delta\varphi \in [0, \pi]$ respectively the energy, angle with respect to the surface normal and azimuthal angle deviation of the reflected/recycled particle. If $\delta\varphi = 0$, the tangential velocity of the recycled/reflected particle is in the same direction as the incident particle, whereas it is in the opposite direction for $\delta\varphi = \pi$.

The TRIM database consists of a total of 84 tables for different combinations of incident energies $E' = 1, 2, 5, 10, 20, 50, 100, 200, 500, 1\,000, 2\,000$ and $5\,000$ eV, and angles $\vartheta' = 0, 30, 45, 60, 70, 80$ and 85 degrees. Every table contains the value for the particle reflection coefficient $R_N(E', \vartheta')$, defined as

$$R_N(E', \vartheta') = \int_{E=0}^{E'} \int_{\cos \vartheta=0}^1 \int_{\cos \delta\varphi=-1}^1 R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta\varphi) \dots dE d(\cos \vartheta) d(\cos \delta\varphi), \quad (3.13)$$

which is the probability that a particle with energy E' and angle ϑ' is reflected as a fast atom.

The particle reflection coefficient $R_N(E', \vartheta')$ only gives the probability of fast reflection, but for the distribution regarding the energy and direction of the recycled/reflected particle, we need extra information. For a certain combination of incident particle energy E' and polar angle ϑ' , three conditional distributions are tabulated: i) $R_F(E', \vartheta' \rightarrow E)$ for the reflected energy distribution (E); ii) $R_F(E', \vartheta' \rightarrow \cos \vartheta|E)$ for the polar angle distribution ($\cos \vartheta$) knowing that the recycled/reflected particle has energy E ; and iii) $R_F(E', \vartheta' \rightarrow \cos \delta\varphi|E, \cos \vartheta)$ for the azimuthal angle distribution ($\cos \delta\varphi$) after we know the recycled/reflected energy E and (cosine of) the polar angle $\cos \vartheta$. These conditional distributions follow from

$$R_F(E', \vartheta' \rightarrow E) = \left(\int_{\cos \vartheta=0}^1 \int_{\cos \delta\varphi=-1}^1 R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \dots \dots \cos \delta\varphi) d(\cos \vartheta) d(\cos \delta\varphi) \right) / R_N(E', \vartheta'), \quad (3.14)$$

$$R_F(E', \vartheta' \rightarrow \cos \vartheta|E) = \left(\int_{\cos \delta\varphi=-1}^1 R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta\varphi) \dots \dots d(\cos \delta\varphi) \right) / (R_N(E', \vartheta') R_F(E', \vartheta' \rightarrow E)), \quad (3.15)$$

$$R_F(E', \vartheta' \rightarrow \cos \delta\varphi|E, \cos \vartheta) = R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta\varphi) / (R_N(E', \vartheta') \dots \dots R_F(E', \vartheta' \rightarrow E) R_F(E', \vartheta' \rightarrow \cos \vartheta|E)). \quad (3.16)$$

Then, the total reflection kernel can be written as

$$R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta\varphi) = R_N(E', \vartheta') R_F(E', \vartheta' \rightarrow E) \dots \dots R_F(E', \vartheta' \rightarrow \cos \vartheta|E) R_F(E', \vartheta' \rightarrow \cos \delta\varphi|E, \cos \vartheta). \quad (3.17)$$

These conditional distributions can be considered as three one-dimensional probability density functions. In the TRIM database, these distributions are discretized by means of five quantiles. We generalize the distribution as $R_F(E', \vartheta' \rightarrow x)$, $x \in [x_{\min}, x_{\max}]$, with $x = E$ for the conditional energy distribution, $x = \cos \vartheta$ for the polar angle distribution, and $x = \cos \delta\varphi$ for the azimuthal angle distribution. Then, the cumulative distribution $F_F(E', \vartheta' \rightarrow x)$ is defined as

$$F_F(E', \vartheta' \rightarrow x) = \int_{x_{\min}}^x R_F(E', \vartheta' \rightarrow y) dy. \quad (3.18)$$

This distribution is discretized with five quantiles:

$$F_F(E', \vartheta' \rightarrow x_i) = q_i, \quad \text{for } i = 1, \dots, 5, \quad (3.19)$$

with $q_i = 0.1 + 0.2(i - 1)$. This is illustrated in Fig. 3.2 (a), where the blue line represents the real cumulative distribution. The linear interpolation between these quantiles leads to a piecewise constant approximation for the probability density functions, as can be seen in Fig. 3.2 (b).

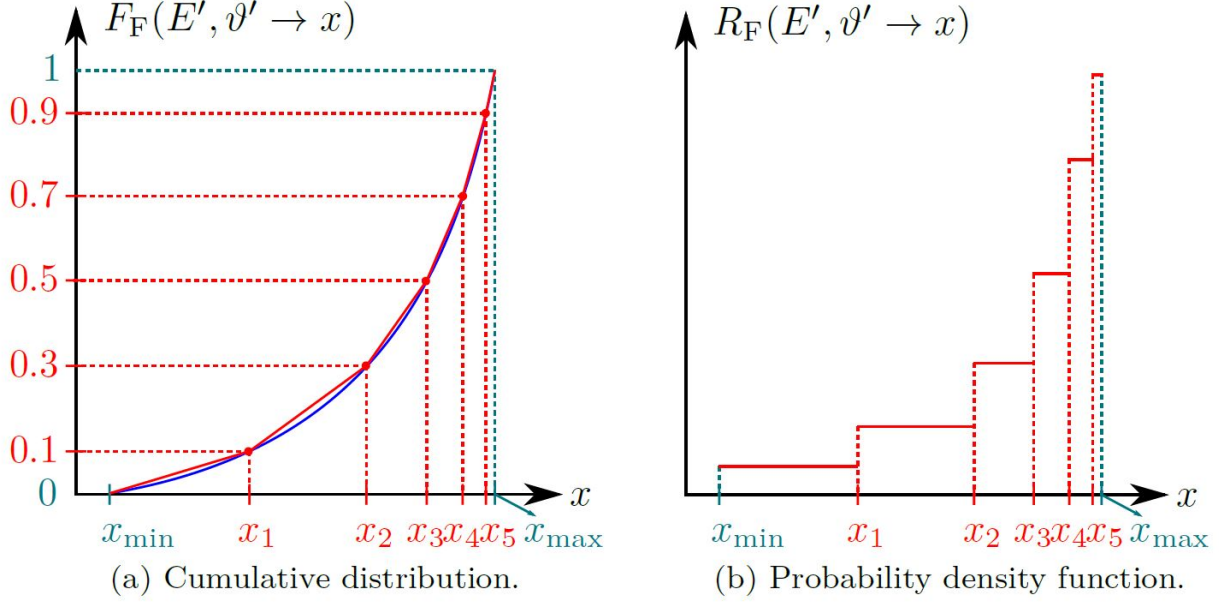


Figure 3.2: Conditional cumulative distribution and probability density function from the TRIM database.

The cumulative distribution $F_F(E', \vartheta' \rightarrow E)$ is discretized with five quantiles E_i for every value of E' and ϑ' . Then for every quantile E_i , the function $F_F(E', \vartheta' \rightarrow \cos \vartheta | E_i)$ is discretized with five quantiles $\cos \vartheta_j$, thus, 25 additional numbers. Finally, the same is done for $F_F(E', \vartheta' \rightarrow \cos \delta \varphi | E_i, \cos \vartheta_j)$ (125 additional numbers). Hence, for every combination of E' and ϑ' the TRIM database contains 155 numbers + 1 additional number for the reflection coefficient. Fig. 3.3 shows an example of a table from the TRIM database for a certain incident energy-angle combination. The last number of the first line is the reflection coefficient $R_N(E', \vartheta')$. The second line contains the quantiles for the reflected energy, the next five lines the quantiles for $\cos \vartheta$ for every value of E , and the next 25 lines contain the quantiles for $\cos \delta \varphi$ for every combination of E and $\cos \vartheta$.

$E'[\text{eV}]$				$\vartheta' [^\circ]$	$R_N(E', \vartheta')$	
1.	2.00	26.	56.00	2.00E+02	3.00E+01	4.80E-01
				$\rho_1 = 0.3 \Rightarrow i = 2$		
4.11620E+01	9.16650E+01	1.25390E+02	1.50840E+02	1.72260E+02	$E_i, i = 1, \dots, 5$	
				$\rho_2 = 0.5 \Rightarrow j = 3$		
4.82510E-01	6.71020E-01	7.93260E-01	8.86500E-01	9.64970E-01	$\cos \vartheta_j^i, j = 1, \dots, 5$	
4.64730E-01	6.53370E-01	7.78110E-01	8.76670E-01	9.60970E-01		
4.33170E-01	6.27780E-01	7.58580E-01	8.67410E-01	9.57180E-01		
4.18910E-01	6.07690E-01	7.37670E-01	8.49380E-01	9.52100E-01		
3.16180E-01	4.77130E-01	6.08050E-01	7.38090E-01	8.90090E-01		
				$\rho_3 = 0.9 \Rightarrow k = 5$		
-9.40740E-01	-5.56890E-01	6.39960E-02	6.32580E-01	9.56660E-01	$i = 1$	$\cos \delta \varphi_k^{i,j}, k = 1, \dots, 5$
-9.43650E-01	-5.83060E-01	-2.24480E-02	5.53250E-01	9.38340E-01		
-9.42090E-01	-5.90000E-01	-3.12500E-02	5.86500E-01	9.51360E-01		
-9.48140E-01	-5.63720E-01	3.37290E-03	6.00100E-01	9.56280E-01		
-9.48830E-01	-5.67360E-01	-1.34340E-02	5.95890E-01	9.56690E-01		
-9.39810E-01	-5.87230E-01	-2.29360E-02	5.59450E-01	9.44810E-01	$i = 2$	
-9.54800E-01	-5.73700E-01	3.88110E-02	5.87850E-01	9.49180E-01		
-9.45890E-01	-5.85990E-01	-4.72020E-02	5.59330E-01	9.42820E-01		
-9.46810E-01	-6.23760E-01	-4.98030E-02	5.61960E-01	9.39650E-01		
-9.52460E-01	-5.42690E-01	5.07270E-02	6.15900E-01	9.60990E-01		
-9.45580E-01	-5.56640E-01	2.38740E-02	6.11950E-01	9.62670E-01	$i = 3$	
-9.42630E-01	-5.42540E-01	3.99670E-02	6.07410E-01	9.51140E-01		
-9.52150E-01	-5.53460E-01	4.96280E-02	6.20170E-01	9.55360E-01		
-9.36370E-01	-5.59120E-01	2.80660E-02	6.24860E-01	9.44440E-01		
-9.48420E-01	-5.61020E-01	3.99420E-03	5.94790E-01	9.49920E-01		
-9.35360E-01	-4.59090E-01	1.55120E-01	6.70960E-01	9.52260E-01	$i = 4$	
-9.36900E-01	-4.98860E-01	1.52800E-01	6.60320E-01	9.60450E-01		
-9.37110E-01	-5.15800E-01	1.07910E-01	6.31570E-01	9.61270E-01		
-9.35620E-01	-5.13910E-01	1.37290E-01	6.88520E-01	9.63320E-01		
-9.43480E-01	-5.41080E-01	7.34930E-02	6.16980E-01	9.47230E-01		
-7.03640E-01	-2.60170E-02	4.62110E-01	8.07130E-01	9.81170E-01	$i = 5$	
-6.50590E-01	-7.77770E-04	5.20590E-01	8.27840E-01	9.80430E-01		
-6.36710E-01	-3.04950E-02	4.52520E-01	7.94420E-01	9.75290E-01		
-6.26120E-01	-5.22300E-02	4.84880E-01	8.20230E-01	9.78380E-01		
-5.59810E-01	-7.71550E-02	3.96120E-01	7.88740E-01	9.76690E-01		

Figure 3.3: Data block from the TRIM database for $E' = 200$ eV and $\vartheta' = 30^\circ$ [17].

From the expression for $R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta \varphi)$ we can derive the reflection kernel $R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi)$ in Eq. (3.3), via the relation

$$R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi) \sin \vartheta dv d\vartheta d\varphi = \frac{1}{2} R_F(E', \vartheta' \rightarrow E, \cos \vartheta, \cos \delta \varphi) \dots \dots dE |d(\cos \vartheta) d(\cos \delta \varphi)|. \quad (3.20)$$

In the right hand side, there appears a factor 1/2, because of the equal probability that the particle is azimuthally deviating in the clockwise or counterclockwise direction compared to its original trajectory. With

$$E' = \frac{m}{2} (v')^2, \quad E = \frac{m}{2} v^2, \quad \varphi = \varphi' + \pi \pm \delta \varphi, \quad (3.21)$$

this becomes

$$R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi) = \frac{m}{2} v |\sin(\varphi - \varphi')| R_F\left(\frac{m}{2}(v')^2, \vartheta' \rightarrow \frac{m}{2}v^2, \cos \vartheta, \cos(\varphi - \varphi')\right). \quad (3.22)$$

3.1.3 Treatment of boundary conditions in Monte Carlo simulation

Application of the TRIM database

The specific format for the TRIM tables facilitates the Monte Carlo simulation for the reflection of particles at a boundary. If a particle hits a boundary, its energy and incident polar angle are calculated. Then, the reflection coefficient R_N is determined by means of linear interpolation or extrapolation of the TRIM data. A random number ρ from a uniform distribution between 0 and 1 is generated. If $\rho \leq R_N$ the particle is reflected as a fast atom and the TRIM database is applied (option 1 below), otherwise the particle is thermally released (option 2 below).

- *Option 1: fast reflection.* Three uniformly distributed random numbers ρ_1 , ρ_2 and ρ_3 (between 0 and 1) are generated. They determine respectively the reflected energy, polar angle and azimuthal angle. The reflected energy follows from

$$F_F(E', \vartheta' \rightarrow E) = \rho_1, \quad (3.23)$$

with the piecewise linear approximation of the cumulative distribution (Fig. 3.2 (a)). Then, the polar angle is determined by

$$F_F(E', \vartheta' \rightarrow \cos \vartheta | E) = \rho_2, \quad (3.24)$$

and finally, the relation for the azimuthal angle is

$$F_F(E', \vartheta' \rightarrow \cos \delta\varphi | E, \cos \vartheta) = \rho_3. \quad (3.25)$$

For the three random numbers $\rho_1 = 0.3$, $\rho_2 = 0.5$ and $\rho_3 = 0.9$ and an incident energy of 200 eV and polar angle of 30° for the target-projectile combination of Fig. 3.3, the resulting reflected energy and angles are indicated in the figure. For intermediate values of E' and ϑ' and random numbers that do not exactly correspond with a tabulated quantile, linear interpolation/extrapolation of the tables is used.

- *Option 2: thermal release.* In case of thermal release, the particle gets an energy of 2-3 eV. In order to have an isotropically distributed surface source (Lambert's cosine law), the polar angle has to be sampled from the probability density function $f(\vartheta)$, given by

$$f(\vartheta) = 2 \cos \vartheta \sin \vartheta, \quad \text{for } \vartheta \in [0, \pi/2], \quad (3.26)$$

as can be seen in Eq. (3.3) [9]. The corresponding cumulative distribution is

$$\begin{aligned} F(\vartheta) &= \int_0^\vartheta f(\vartheta') d\vartheta' \\ &= \sin^2 \vartheta. \end{aligned} \quad (3.27)$$

Then, the polar angle follows from the relation

$$F(\vartheta) = \rho, \quad (3.28)$$

with ρ a random number uniformly distributed between 0 and 1. This leads to

$$\vartheta = \arcsin(\sqrt{\rho}). \quad (3.29)$$

The azimuthal reflected angle is uniformly sampled between 0 and 2π .

3.2 Fluid boundary conditions

In this section, we derive the boundary conditions for the fluid neutral models. The fluid models as described in literature use simple boundary conditions. E.g., a typical boundary condition for the parallel momentum equation is the momentum recycling boundary condition where the neutral parallel velocity at the target plate is assumed to be a constant fraction of the ion parallel velocity [18, 19]. However, this fraction is user-defined and not based on the underlying physics.

Here, we carefully derive the boundary conditions from the underlying kinetic physics. This is done by taking the particular moments of the neutral velocity distribution at the boundary. The total distribution consists of three parts: the incident neutrals, the reflected neutrals (an incident atom that is reflected as a fast or thermal atom) and the recycled neutrals (the incident ions that pick up an electron and are emitted as fast or thermal atoms). The distribution of the recycled neutrals can be calculated by taking the ion distribution at the boundary that entirely follows from the plasma macroscopic properties (Eq. (3.11)), and multiplying it by the reflection kernel and integrating this product over the incident velocity space (Eqs. (3.3)-(3.4)). The story is different for the reflected neutrals, because we need an expression for the distribution of the incident neutrals, but this can only be known by solving the kinetic equation. For the fluid models, we make an estimate for the distribution of these incident atoms.

The estimate is based on the diffusion approximation, as explained in Section 3.2.1, or the drifting Maxwellian approximation, as explained in Section 3.2.2. Again, applying the reflection kernel leads to an estimate for the distribution of the reflected particles. In Section 3.2.3, we take moments from the total distribution to obtain macroscopic fluxes. Finally, Section 3.2.6 deals with the practical implementation of these boundary conditions with imposed fluxes. It should be noted that the error introduced by the boundary conditions is fully due to the approximation for the incident neutrals. The TRIM reflection model is used for both the fluid and kinetic models and consequently, it does not introduce additional errors.

3.2.1 Diffusion approximation for the distribution of the incident neutrals

Diffusion theory is frequently used in nuclear (fission) reactor physics [8, 14, 15, 16]. In our case, we will exclusively use it to estimate the distribution of the incident neutrals. We consider a control volume dV in which charge-exchange collisions scatter the neutrals in different directions, as indicated in Fig. 3.4. We are interested in the number of neutrals that reach the area dA spanned by the solid angle $d\Omega$. dA is a surface element of the boundary (e.g., the target plate).

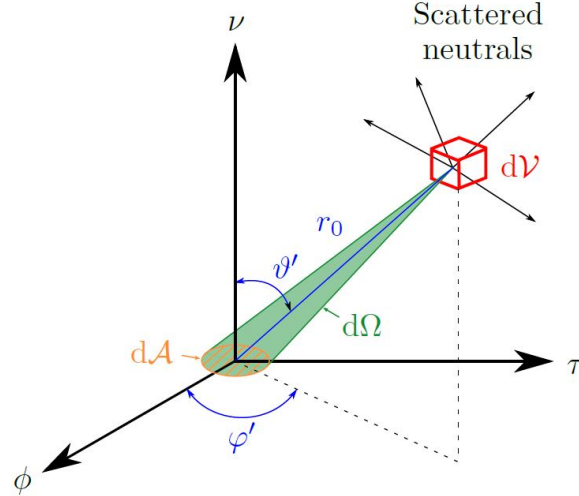


Figure 3.4: Scattering of neutrals in control volume dV [12].

We calculate the speed- and angular-dependent particle flux density $\Gamma_{\nu-}^n(v', \vartheta', \varphi')$ of neutrals scattered by the ions and reaching the plane $d\mathcal{A}$. We interpret $\Gamma_{\nu-}^n(v', \vartheta', \varphi')dv'd\vartheta'd\varphi'$ as the number of neutrals per m^2 per second reaching the surface element $d\mathcal{A}$ with a polar angle between ϑ' and $\vartheta' + d\vartheta'$, an azimuthal angle between φ' and $\varphi' + d\varphi'$, and a speed between v' and $v' + dv'$.

To calculate $\Gamma_{\nu-}^n(v', \vartheta', \varphi')$, we need the source $S(v', \vartheta', \varphi')$ of neutrals created in the volume element $d\mathcal{V}$ with a speed between v' and $v' + dv'$ scattered in the solid angle $d\Omega$. Charge-exchange collisions are assumed to be the main origin for neutral particles at a certain distance r_0 . Thus, the source can be written as

$$S(v', \vartheta', \varphi') = n_n n_i K_{\text{cx},m} \tilde{f}_i(v', \vartheta', \varphi'), \quad (3.30)$$

which is the number of charge-exchange collisions per unit volume per second, multiplied with the normalized ion distribution $\tilde{f}_i(v', \vartheta', \varphi')$. Here, n_n is the neutral particle density, n_i is the ion particle density, and $K_{\text{cx},m}$ is the momentum-linearized charge-exchange rate, as defined in Ref. [12].

The ion distribution in the domain is assumed to be a drifting Maxwellian ($\tilde{f}_i(\mathbf{v}') = \tilde{M}(\mathbf{v}'; \mathbf{V}_i, T_i)$). This ion distribution is expressed as a function of the three velocity components and has to be transformed to the speed- and angular-dependent distribution as a function of v' , ϑ' and φ' . This is done with the transformation from Cartesian to spherical coordinates:

$$\tilde{f}_i(\mathbf{v}')d\mathbf{v}' = \tilde{f}_i(\mathbf{v}')(v')^2 dv'd\Omega = \tilde{f}_i(v', \vartheta', \varphi')dv'd\Omega. \quad (3.31)$$

In this way, $\tilde{f}_i(v', \vartheta', \varphi')$ becomes

$$\begin{aligned} \tilde{f}_i(v', \vartheta', \varphi') &= (v')^2 \tilde{M}(\mathbf{v}'; \mathbf{V}_i, T_i) \\ &= (v')^2 \left(\frac{m}{2\pi T_i} \right)^{3/2} \exp \left(-\frac{m}{2T_i} ((v'_\phi - u_\phi)^2 + (v'_\tau - u_\tau)^2 \right. \\ &\quad \left. + (v'_\nu - u_\nu)^2) \right). \end{aligned} \quad (3.32)$$

Also transforming the Cartesian particle velocities to spherical coordinates, gives

$$\begin{aligned} \tilde{f}_i(v', \vartheta', \varphi') &= (v')^2 \left(\frac{m}{2\pi T_i} \right)^{3/2} \exp \left(-\frac{m}{2T_i} \left((v')^2 + 2v' \sin \vartheta' \sin \varphi' u_\tau \right. \right. \\ &\quad \left. \left. + 2v' \sin \vartheta' \cos \varphi' u_\phi + 2v' \cos \vartheta' u_\nu + u_\phi^2 + u_\tau^2 + u_\nu^2 \right) \right). \end{aligned} \quad (3.33)$$

This source has to be multiplied with the probability that the neutral reaches the boundary plane without any extra collision (probability of non-absorption $P_{\text{NA}}(v')$). This probability is given by

$$P_{\text{NA}}(v') = \exp(-\Sigma_t(v')r_0), \quad (3.34)$$

with $\Sigma_t(v') = (n_i K_{\text{cx},m} + n_e K_i)/v'$ the total macroscopic cross-section (including charge-exchange and ionization events).

Then, the speed- and angular-dependent particle flux density becomes

$$\Gamma_{\nu-}^n(v', \vartheta', \varphi')dv'd\vartheta'd\varphi'd\mathcal{A} = \int_{\mathcal{V}} S(v', \vartheta', \varphi')P_{\text{NA}}(v')d\mathcal{V}dv'd\Omega. \quad (3.35)$$

To obtain the total particle flux density, the integral over the whole domain volume has to be taken. If the total macroscopic cross-section is large, the exponential term $e^{-\Sigma_t(v')r_0}$ in $P_{\text{NA}}(v')$ (Eq. (3.34)) decays rapidly. Therefore, only the first few mean free paths starting from the boundary contribute to the integral of Eq. (3.35), and we can replace the integral over the edge plasma domain by a semi-infinite integral with

$r_0 \in [0, \infty)$. The solid angle spanned by $d\mathcal{A}$ is $d\mathcal{A} \frac{\cos \vartheta'}{r_0^2}$ and the volume $d\mathcal{V}$ is $r_0^2 \sin \vartheta' dr_0 d\vartheta' d\varphi'$. Inserting all expressions in Eq. (3.35), gives

$$\Gamma_{\nu-}^n(v', \vartheta', \varphi') = \int_{r_0=0}^{\infty} e^{-\Sigma_t(v')r_0} n_n n_i K_{\text{cx,m}} \tilde{f}_i(v', \vartheta', \varphi') \sin \vartheta' \cos \vartheta' dr_0. \quad (3.36)$$

We use a Taylor series expansion for the neutral density:

$$\begin{aligned} n_n &\approx n_{n0} + \phi \left(\frac{\partial n_n}{\partial \phi} \right)_0 + \tau \left(\frac{\partial n_n}{\partial \tau} \right)_0 + \nu \left(\frac{\partial n_n}{\partial \nu} \right)_0 \\ &= n_{n0} + r_0 \sin \vartheta' \sin \varphi' \left(\frac{\partial n_n}{\partial \tau} \right)_0 + r_0 \cos \vartheta' \left(\frac{\partial n_n}{\partial \nu} \right)_0, \end{aligned} \quad (3.37)$$

assuming toroidal symmetry such that $(\partial n_n / \partial \phi)_0 = 0$. The subscript 0 indicates that the neutral density and gradients are evaluated in the origin, where the particle flux density is calculated. Inserting this expression for the neutral density in Eq. (3.36), gives

$$\begin{aligned} \Gamma_{\nu-}^n(v', \vartheta', \varphi') &= \int_{r_0=0}^{\infty} e^{-\Sigma_t(v')r_0} \left(n_n + r_0 \sin \vartheta' \sin \varphi' \left(\frac{\partial n_n}{\partial \tau} \right) \right. \\ &\quad \left. + r_0 \cos \vartheta' \left(\frac{\partial n_n}{\partial \nu} \right) \right) n_i K_{\text{cx,m}} \tilde{f}_i(v', \vartheta', \varphi') \sin \vartheta' \cos \vartheta' dr_0, \end{aligned} \quad (3.38)$$

where the subscript 0 is omitted for simplicity of notation. Finally, evaluating the integral neglecting the spatial dependence of Σ_t , $n_i K_{\text{cx,m}}$ and $\tilde{f}_i$, gives

$$\begin{aligned} \Gamma_{\nu-}^n(v', \vartheta', \varphi') &= \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \tilde{f}_i(v', \vartheta', \varphi') v' \sin \vartheta' \cos \vartheta' \left(n_n \right. \\ &\quad \left. + \frac{v'}{n_i K_{\text{cx,m}} + n_e K_i} \left(\sin \vartheta' \sin \varphi' \frac{\partial n_n}{\partial \tau} + \cos \vartheta' \frac{\partial n_n}{\partial \nu} \right) \right). \end{aligned} \quad (3.39)$$

Due to the exponential decay term in Eq. (3.38), only the first mean free paths away from the boundary contribute to the integral and the spatial variation of Σ_t , $n_i K_{\text{cx,m}}$ and $\tilde{f}_i$ has to be small in this region. Basically, this means that the Knudsen number has to be sufficiently small, which is also a necessary condition for the use of a fluid model in the bulk of the domain. Indeed, a small Knudsen number means that the macroscopic length scales of the parameters in the integrand of Eq. (3.38) are large compared to the neutral mean free path length ($1/\Sigma_t$). These large macroscopic length scales compared to the mean free path length lead to relatively small variations of the macroscopic properties in the exponentially decaying region that contributes to the integral.

3.2.2 Truncated drifting Maxwellian approximation for the distribution of the incident neutrals

The diffusion approximation for the incident neutrals was shown to be accurate for the target boundary conditions in [12] and [2]. However, this boundary condition requires a short mean free path to be valid, and it was found that unphysical behavior can arise when the mean free path is too long. Therefore, an alternative model was developed, assuming a truncated drifting Maxwellian distribution for the incident neutrals, based on the macroscopic neutral density, velocity and temperature at the target:

$$\Gamma_{\nu-}^n(v', \vartheta', \varphi') = \tilde{f}_n(v', \vartheta', \varphi') v' \sin \vartheta' \cos \vartheta'. \quad (3.40)$$

Here, the normalized neutral distribution is

$$\begin{aligned}\tilde{f}_n(v', \vartheta', \varphi') &= (v')^2 \tilde{M}(\mathbf{v}'; \mathbf{V}_n, T_n) \\ &= (v')^2 \left(\frac{m}{2\pi T_n} \right)^{3/2} \exp \left(-\frac{m}{2T_n} ((v'_\phi - u_{n\phi})^2 + (v'_\tau - u_{n\tau})^2 \right. \\ &\quad \left. + (v'_\nu - u_{n\nu})^2) \right).\end{aligned}\quad (3.41)$$

Again transforming the Cartesian particle velocities to spherical coordinates, gives

$$\begin{aligned}\tilde{f}_n(v', \vartheta', \varphi') &= (v')^2 \left(\frac{m}{2\pi T_n} \right)^{3/2} \exp \left(-\frac{m}{2T_n} \left((v')^2 + 2v' \sin \vartheta' \sin \varphi' u_{n\tau} \right. \right. \\ &\quad \left. \left. + 2v' \sin \vartheta' \cos \varphi' u_{n\phi} + 2v' \cos \vartheta' u_{n\nu} + u_{n\phi}^2 + u_{n\tau}^2 + u_{n\nu}^2 \right) \right).\end{aligned}\quad (3.42)$$

This option was found to be slightly less accurate than the diffusion approximation for high collisional cases [26], but still far more accurate than previous ad-hoc fluid neutral boundary conditions, while being more robust in the regions with larger neutral mean free paths. From now on, we use the superscripts ‘diff’ and ‘maxw’ to differentiate between diffusion or Maxwellian option, when the distinction needs to be made explicit.

3.2.3 Resulting macroscopic boundary fluxes

From the distributions of the incident neutrals and ions, the net fluxes at the boundaries can be calculated. In our case, the speed and angular distributions of the fast recycled and reflected neutrals are determined by the TRIM code database, whereas the thermally released neutrals are emitted diffusely with an energy of 2-3 eV. Below, we elaborate the particle, parallel momentum and energy flux density.

Neutral particle flux density

The net neutral particle flux density $\Gamma^n = n_n \mathbf{V}_n \cdot \boldsymbol{\nu}$ constitutes of four contributions:

$$\Gamma^n = \Gamma_{\text{refl},f}^n + \Gamma_{\text{rec},f}^n + \Gamma_T^n - \Gamma_{\nu-}^n, \quad (3.43)$$

with $\Gamma_{\text{refl},f}^n$ and $\Gamma_{\text{rec},f}^n$ respectively the fast reflected and recycled particle flux density (as described by the TRIM database) and Γ_T^n the thermally released fraction, decreased with the incident particle flux density $\Gamma_{\nu-}^n$. These contributions are elaborated with the help of the particle reflection coefficient (Eq. (3.13)):

$$\Gamma_{\nu-}^n = \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \Gamma_{\nu-}^n(v', \vartheta', \varphi') dv' d\vartheta' d\varphi', \quad (3.44)$$

$$\Gamma_{\text{refl},F}^n = R^n \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta') \Gamma_{\nu-}^n(v', \vartheta', \varphi') dv' d\vartheta' d\varphi', \quad (3.45)$$

$$\Gamma_{\text{rec},F}^n = R^i \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta') \Gamma_{\nu-}^i(v', \vartheta', \varphi') dv' d\vartheta' d\varphi', \quad (3.46)$$

$$\begin{aligned}\Gamma_T^n &= \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} (1 - R_N(v', \vartheta')) (R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') \\ &\quad + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi')) dv' d\vartheta' d\varphi'.\end{aligned}\quad (3.47)$$

It should be noted that the particle reflection coefficient is written as a function of the incident speed v' and not as function of the incident energy $E' = \frac{m}{2}(v')^2$ in Eq. (3.13).

Neutral parallel momentum flux density

Similar to the net neutral particle flux density, the neutral parallel momentum flux density $\Gamma_{m,||}^n$ becomes

$$\begin{aligned}
\Gamma_{m,||}^n &= [(mn_n \mathbf{V}_n \mathbf{V}_n + \Pi_n + p_n \mathbb{I}) \cdot \boldsymbol{\nu}]_{||} \\
&= m \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \left[\underbrace{-v'_{||} \Gamma_{\nu-}^n(v', \vartheta', \varphi')}_{\text{incident}} \right. \\
&\quad + \underbrace{\int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} v_{||} R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi) \sin \vartheta dv d\vartheta d\varphi \dots}_{\text{fast}} \\
&\quad \left. \dots \underbrace{(R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi'))}_{\text{fast}} \right] dv' d\vartheta' d\varphi' + \underbrace{\Gamma_{m,||,T}^n}_{\text{thermal}}, \tag{3.48}
\end{aligned}$$

with $v_{||} = b_\theta(\sin \alpha_\nu v_\tau + \cos \alpha_\nu v_\nu) + b_\phi v_\phi$ the neutral particle velocity parallel to the magnetic field, with α_ν the angle between the surface normal and the poloidal direction. The different contributions are indicated with sub braces. To write Eq. (3.49) more compactly, we make use of the parallel and perpendicular momentum reflection coefficients, respectively $R_{m,||}(v', \vartheta')$ and $R_{m,\perp}(v', \vartheta')$. Then, Eq. (3.49) becomes

$$\begin{aligned}
\Gamma_{m,||}^n &= m \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \left[\underbrace{-v'_{||} \Gamma_{\nu-}^n(v', \vartheta', \varphi')}_{\text{incident}} + \underbrace{\left(-b_\phi R_{m,||}(v', \vartheta') \dots \right)}_{\text{fast}} \right. \\
&\quad \underbrace{\dots \sin \vartheta' \cos \varphi' - b_\theta \sin \alpha_\nu R_{m,||}(v', \vartheta') \sin \vartheta' \sin \varphi' + b_\theta \cos \alpha_\nu \dots}_{\text{fast}} \\
&\quad \left. \underbrace{\dots R_{m,\perp}(v', \vartheta') \cos \vartheta'}_{\text{fast}} \right) v' (R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi')) \dots \\
&\quad \dots dv' d\vartheta' d\varphi' + \underbrace{\Gamma_{m,||,T}^n}_{\text{thermal}}, \tag{3.50}
\end{aligned}$$

with

$$\begin{aligned}
R_{m,||}(v', \vartheta') &= \frac{1}{mv' \sin \vartheta'} \int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} mv \sin \vartheta \cos \varphi \dots \\
&\quad \dots R_F(v', \vartheta', 0 \rightarrow v, \vartheta, \varphi) \sin \vartheta dv d\vartheta d\varphi, \tag{3.51}
\end{aligned}$$

$$\begin{aligned}
R_{m,\perp}(v', \vartheta') &= \frac{1}{mv' \cos \vartheta'} \int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} mv \cos \vartheta R_F(v', \vartheta', 0 \rightarrow v, \vartheta, \varphi) \dots \\
&\quad \dots \sin \vartheta dv d\vartheta d\varphi. \tag{3.52}
\end{aligned}$$

The parallel momentum reflection coefficient represents the mean fraction of momentum that is conserved in the direction of motion of the incident particle projected on the boundary plane multiplied by the probability of fast reflection $R_N(v', \vartheta')$. The perpendicular momentum reflection coefficient represents the ratio of the momentum perpendicular to the plane of the reflected and incident particle again multiplied by $R_N(v', \vartheta')$. The calculation of these reflection coefficients is treated in Section 3.2.6.

The thermally released neutrals are emitted isotropically (Lambert's cosine law), leading to the following

parallel momentum flux density:

$$\begin{aligned}\Gamma_{m,||,T}^n &= m \int_{v=0}^{\infty} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} v_{||} \frac{1}{\pi} \delta(v - v_T) \cos \vartheta \sin \vartheta \Gamma_T^n \\ &= b_\theta \cos \alpha_\nu \frac{2}{3} m v_T \Gamma_T^n.\end{aligned}\quad (3.53)$$

As can be seen in Eq. (3.48), this parallel momentum flux density includes the pressure. However, for fluid models it is typical to consider the pressure gradient as a driving force rather than a flux contribution. The parallel momentum flux density from the fluid point of view (subscript f) becomes

$$\Gamma_{m,f,||}^n = [\Gamma_{m,f}^n \cdot \boldsymbol{\nu}]_{||} = \Gamma_{m,||}^n - b_\theta \cos \alpha_\nu p_n, \quad (3.54)$$

i.e., excluding the pressure contribution.

There are two possible choices for the pressure at a certain position at a boundary. One can take the pressure that follows from the fluid solution. However, at the boundaries the underlying velocity distribution deviates a lot from the Maxwellian equilibrium distribution. Therefore, it is recommended to recalculate the pressure from the estimated distribution that consists of the incident, recycled and reflected neutrals. If the total velocity-dependent particle flux density at a boundary is $\Gamma^n(\mathbf{v})$, the pressure is defined as

$$p'_n = \frac{m}{3} \int ||\mathbf{v} - \mathbf{V}'_n||^2 \frac{\Gamma^n(\mathbf{v})}{|v_\nu|} d\mathbf{v}, \quad (3.55)$$

where the prime is used to indicate that the macroscopic quantity is recalculated from the estimated underlying velocity distribution $\Gamma^n(\mathbf{v})/|v_\nu|$ at the boundary, and consequently, it might differ from the quantity that results from the fluid solution. This is not only the case for the pressure, but also for the density ($n'_n = \int \frac{\Gamma^n(\mathbf{v})}{|v_\nu|} d\mathbf{v}$) and velocity ($\mathbf{V}'_n = \frac{1}{n'_n} \int \mathbf{v} \frac{\Gamma^n(\mathbf{v})}{|v_\nu|} d\mathbf{v}$).

Elaborating Eq. (3.55), gives

$$p'_n = \underbrace{\frac{m}{3} \int ||\mathbf{v}||^2 \frac{\Gamma^n(\mathbf{v})}{|v_\nu|} d\mathbf{v}}_{(1)} - \underbrace{\frac{m}{3} n'_n ||\mathbf{V}'_n||^2}_{(2)}. \quad (3.56)$$

Term (1) is the total second order moment with a contribution from the fluid velocity $\mathbf{V}'_n$ that has to be subtracted to obtain the pressure (term (2)).

Elaborating term (1) of Eq. (3.56), gives

$$\begin{aligned}\frac{m}{3} \int ||\mathbf{v}||^2 \frac{\Gamma^n(\mathbf{v})}{|v_\nu|} d\mathbf{v} &= \frac{m}{3} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \left[\underbrace{(v')^2 \frac{\Gamma_{\nu-}^n(v', \vartheta', \varphi')}{v' \cos \vartheta'}}_{\text{incident}} \right. \\ &\quad \left. + \underbrace{R_p(v', \vartheta') (v')^2 \frac{R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi')}{v' \cos \vartheta'}}_{\text{fast}} \right] dv' d\vartheta' d\varphi' + \underbrace{\frac{2}{3} m v_T \Gamma_T^n}_{\text{thermal}},\end{aligned}\quad (3.57)$$

with

$$\begin{aligned}R_p(v', \vartheta') &= \frac{\cos \vartheta'}{v'} \int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} \frac{v}{\cos \vartheta} R_F(v', \vartheta', 0 \rightarrow v, \vartheta, \varphi) \sin \vartheta \dots \\ &\dots dv d\vartheta d\varphi,\end{aligned}\quad (3.58)$$

a reflection coefficient that determines the second order moment of the distribution after reflection.

Now, we elaborate term (2) of Eq. (3.56). Therefore, we need an expression for the density n'_n and the velocity $\mathbf{V}'_n$. To calculate these quantities, we need the distributions of the different contributions, i.e., the distribution of the incident, fast reflected/recycled and thermally released neutrals, respectively $f_{n,\nu-}(v', \vartheta', \varphi')$, $f_{n,\nu+,f}(v, \vartheta, \varphi)$ and $f_{n,\nu+,T}(v, \vartheta, \varphi)$, given by

$$f_{n,\nu-}(v', \vartheta', \varphi') = \frac{\Gamma_{\nu-}^n(v', \vartheta', \varphi')}{v' \cos \vartheta' \sin \vartheta'}, \quad (3.59)$$

$$f_{n,\nu+,f}(v, \vartheta, \varphi) = \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi) \sin \vartheta \dots \\ \dots \frac{R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi')}{v \cos \vartheta \sin \vartheta} dv' d\vartheta' d\varphi', \quad (3.60)$$

$$f_{n,\nu+,T}(v, \vartheta, \varphi) = \frac{\Gamma_T^n}{\pi v_T} \delta(v - v_T). \quad (3.61)$$

The zeroth order moment of the total distribution leads to an expression for n'_n :

$$n'_n = \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} f_{n,\nu-}(v', \vartheta', \varphi') \sin \vartheta' dv' d\vartheta' d\varphi' \\ + \int_{v=0}^{\infty} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} (f_{n,\nu+,f}(v, \vartheta, \varphi) + f_{n,\nu+,T}(v, \vartheta, \varphi)) \sin \vartheta dv d\vartheta d\varphi \\ = \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \left[\frac{\Gamma_{\nu-}^n(v', \vartheta', \varphi')}{v' \cos \vartheta'} + \frac{R_{n_n}(v', \vartheta')}{v' \cos \vartheta'} (R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi')) \right. \\ \left. + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi') \right] dv' d\vartheta' d\varphi' + \frac{2}{v_T} \Gamma_T^n, \quad (3.62)$$

with the density reflection coefficient $R_{n_n}(v', \vartheta')$, defined as

$$R_{n_n}(v', \vartheta') = v' \cos \vartheta' \int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} \frac{1}{v \cos \vartheta} R_F(v', \vartheta', 0 \rightarrow v, \vartheta, \varphi) \sin \vartheta \dots \\ dv d\vartheta d\varphi. \quad (3.63)$$

This coefficient guarantees the particle balance of the recycled/reflected neutrals. If there is a strong reduction of the perpendicular velocity after reflection, the density has to increase to guarantee that the reflected/recycled particle flux density is the incident particle flux density multiplied by the probability of fast reflection.

Finally, we need an expression for $\mathbf{V}'_n$, the last missing factor to calculate the pressure with Eq. (3.56). Assuming that $u'_{n\phi} \gg u'_{n\tau}, u'_{n\nu}$, gives

$$\frac{m}{3} n'_n \|\mathbf{V}'_n\|^2 \approx \frac{m}{3} n'_n u'^2_{n\phi}, \quad (3.64)$$

such that we only have to calculate the toroidal component $u'_{n\phi}$ from the underlying estimated distribution. This assumption is in accordance to the ones made for the parallel momentum equation [12], where it is assumed that the neutral parallel velocity is at least an order of magnitude larger than the radial and diamagnetic components. For small pitches, the toroidal and parallel components are almost the same. The

toroidal velocity follows from

$$\begin{aligned}
n'_n u'_{n\phi} = & \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \underbrace{\left[-v' \sin \vartheta' \cos \varphi' \frac{\Gamma_{\nu-}^n(v', \vartheta', \varphi')}{v' \cos \vartheta'} \right]}_{\text{incident}} \\
& + \underbrace{\int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} v \sin \vartheta \cos \varphi R_F(v', \vartheta', \varphi' \rightarrow v, \vartheta, \varphi) \sin \vartheta dv d\vartheta d\varphi \dots}_{\text{fast}} \\
& \dots \underbrace{\left[\frac{R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi')}{v \cos \vartheta} \right]}_{\text{fast}} dv' d\vartheta' d\varphi'. \quad (3.65)
\end{aligned}$$

There is no contribution from the thermal neutrals to $u'_{n\phi}$ due to the isotropic emission. Again, Eq. (3.65) can be simplified by defining a reflection coefficient for the inner integrals over v , ϑ and φ :

$$\begin{aligned}
R_{u_{n\phi}}(v', \vartheta') = & \frac{1}{\tan \vartheta'} \int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} \tan \vartheta \cos \varphi R_F(v', \vartheta', 0 \rightarrow v, \vartheta, \varphi) \sin \vartheta \dots \\
& \dots dv d\vartheta d\varphi, \quad (3.66)
\end{aligned}$$

leading to

$$\begin{aligned}
n'_n u'_{n\phi} = & \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \left[\underbrace{-\tan \vartheta' \cos \varphi' \Gamma_{\nu-}^n(v', \vartheta', \varphi')}_{\text{incident}} \right. \\
& \left. - \underbrace{R_{u_{n\phi}}(v', \vartheta') \tan \vartheta' \cos \varphi' (R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi'))}_{\text{fast}} \right] dv' d\vartheta' d\varphi'. \quad (3.67)
\end{aligned}$$

The reflection coefficient $R_{u_{n\phi}}(v', \vartheta')$ represents the ratio of the toroidal and perpendicular velocity of the reflected particle divided by the same velocity ratio of the incident particle (multiplied by $R_N(v', \vartheta')$).

Neutral energy flux density

The net neutral energy flux density $Q^n = ((\frac{5}{2}T_n + \frac{m}{2}||\mathbf{V}_n||^2) n_n \mathbf{V}_n + \Pi_n \cdot \mathbf{V}_n + \mathbf{q}_n) \cdot \boldsymbol{\nu}$ becomes

$$\begin{aligned}
Q^n = & \frac{m}{2} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \left[\underbrace{-(v')^2 \Gamma_{\nu-}^n(v', \vartheta', \varphi')}_{\text{incident}} \right. \\
& \left. + \underbrace{R_E(v', \vartheta') (v')^2 (R^n \Gamma_{\nu-}^n(v', \vartheta', \varphi') + R^i \Gamma_{\nu-}^i(v', \vartheta', \varphi'))}_{\text{fast}} \right] dv' d\vartheta' d\varphi' + \underbrace{Q_T^n}_{\text{thermal}}, \quad (3.68)
\end{aligned}$$

making use of the energy reflection coefficient $R_E(v', \vartheta')$, defined as

$$\begin{aligned}
R_E(v', \vartheta') = & \frac{1}{m (v')^2 / 2} \int_{v=0}^{v'} \int_{\vartheta=0}^{\pi/2} \int_{\varphi=0}^{2\pi} \frac{m}{2} v^2 R_F(v', \vartheta', 0 \rightarrow v, \vartheta, \varphi) \dots \\
& \dots \sin \vartheta dv d\vartheta d\varphi. \quad (3.69)
\end{aligned}$$

The energy flux density of the thermally released neutrals becomes

$$Q_T^n = \frac{m}{2} v_T^2 \Gamma_T^n. \quad (3.70)$$

SOLPS-ITER uses internal energy equations instead of total energy equations. The necessary conversion to internal heat boundary fluxes ($\Gamma_{H,i}^n$) follows from the definitions of the energy and heat fluxes [6], resulting in

$$\Gamma_{H,\theta}^n = Q_\theta^n - T_n n_n u_{n||} b_\theta - \frac{m}{2} u_{n||}^2 \Gamma_\theta^n + \frac{\eta^n}{2} \nabla_\theta (u_{n||}^2) \quad (3.71)$$

and

$$\Gamma_{H,r}^n = Q_r^n - \frac{m}{2} u_{n||}^2 \Gamma_r^n + \frac{\eta^n}{2} \nabla_r (u_{n||}^2). \quad (3.72)$$

3.2.4 Automated selection of boundary condition type

The optimal choice between the use of $\Gamma_{\nu-}^{n,\text{diff}}$ and $\Gamma_{\nu-}^{n,\text{maxw}}$ depends on the local Knudsen number. Therefore, an option is implemented to automatically switch between both options in function of the local Kn. For each moment μ we have:

$$\Gamma_\mu^n = \begin{cases} \Gamma_\mu^{n,\text{diff}}, & \text{if } \text{Kn} \leq \text{Kn}^{\text{bd}} \\ \Gamma_\mu^{n,\text{maxw}}, & \text{if } \text{Kn} \geq \text{Kn}^{\text{bm}} \\ \frac{\text{Kn} - \text{Kn}^{\text{bd}}}{\text{Kn}^{\text{bm}} - \text{Kn}^{\text{bd}}} \Gamma_\mu^{n,\text{maxw}} + \frac{\text{Kn}^{\text{bm}} - \text{Kn}}{\text{Kn}^{\text{bm}} - \text{Kn}^{\text{bd}}} \Gamma_\mu^{n,\text{diff}}, & \text{otherwise,} \end{cases} \quad (3.73)$$

where Kn^{bd} and Kn^{bm} are user-defined transition criteria (superscript “bd” = “boundary diffusion”, and superscript “bm” = “boundary Maxwellian”). Linear interpolation between Kn^{bd} and Kn^{bm} is used to prevent discontinuities. Kn is based on only the CX collisions, hence

$$\text{Kn} = \frac{\lambda_{\text{cx}}}{L}, \quad (3.74)$$

with

$$\lambda_{\text{cx}} = \frac{c_{n,th}}{n_i K_{\text{cx},m}}, \quad (3.75)$$

where $c_{n,th}$ is the atom thermal speed. L is a user-defined representative macroscopic length scale of the problem.

3.2.5 Consistency with pumped fraction in EIRENE

Up till now, it was assumed that the particle reflection coefficients R^i and R^n are independent constants. However, present-day EIRENE assumes that if $0 < R^n < 1$ or $0 < R^i < 1$, first the thermally released particles are pumped away. Fast particles are only pumped if the pumped fraction is larger than the probability of thermal release. Furthermore, there is a subtle difference between the treatment on the fluid and kinetic side. On the kinetic side, the correction for absorption is done on a per-particle basis. On the fluid side, the correction for absorption will be done after the integration over velocity space. However, the absorption coefficient will often be small (a few percent), and will then often be smaller than the probability for thermal release for each particle. In that case, the correction for absorption is identical for the fluid and the kinetic treatment. To clarify that the correction for absorption is not fully identical on the fluid and kinetic side, R_{fl}^i and R_{fl}^n are introduced as the effective fluid recycling/reflection coefficients.

First, $p_{\text{rec},F}$ and $p_{\text{ref},F}$ are defined as the fraction of incident particles that are fast recycled or reflected, if there would be no absorption ($R^i = R_{\text{fl}}^i = R^n = R_{\text{fl}}^n = 1$):

$$p_{\text{rec},F} = \frac{\int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta') \Gamma_{\nu-}^i(v', \vartheta', \varphi') dv' d\vartheta' d\varphi'}{\int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \Gamma_{\nu-}^i(v', \vartheta', \varphi') dv' d\vartheta' d\varphi'}, \quad (3.76)$$

and

$$p_{\text{refl},F} = \frac{\int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta') \Gamma_{\nu-}^n(v', \vartheta', \varphi') dv' d\vartheta' d\varphi'}{\int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} \Gamma_{\nu-}^n(v', \vartheta', \varphi') dv' d\vartheta' d\varphi'}. \quad (3.77)$$

Now, if the effective fluid recycling coefficient is smaller than the fast recycled or reflected fraction, we have to reduce the fast recycled/reflected part according to this ratio. Otherwise, no correction is needed. This is formalized by setting

$$p_{\text{rec},F}^{\text{corr}} = \frac{\min(p_{\text{rec},F}, R_{\text{fl}}^i)}{p_{\text{rec},F}}, \quad (3.78)$$

and

$$p_{\text{refl},F}^{\text{corr}} = \frac{\min(p_{\text{refl},F}, R_{\text{fl}}^n)}{p_{\text{refl},F}}. \quad (3.79)$$

The absorption-corrected thermally released fractions $p_{\text{rec},T}$ and $p_{\text{refl},T}$ are then given by

$$p_{\text{rec},T} = \max(R_{\text{fl}}^i - p_{\text{rec},F}, 0), \quad (3.80)$$

and

$$p_{\text{refl},T} = \max(R_{\text{fl}}^n - p_{\text{refl},F}, 0). \quad (3.81)$$

Now we can write the net neutral boundary particle flux in the form consistent with the pumping mechanism in EIRENE. Equations (3.45)-(3.47) become, respectively,

$$\Gamma_{\text{refl},F}^n = p_{\text{refl},F}^{\text{corr}} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta') \Gamma_{\nu-}^n(v', \vartheta', \varphi') dv' d\vartheta' d\varphi', \quad (3.82)$$

$$\Gamma_{\text{rec},F}^n = p_{\text{rec},F}^{\text{corr}} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta') \Gamma_{\nu-}^i(v', \vartheta', \varphi') dv' d\vartheta' d\varphi', \quad (3.83)$$

$$\begin{aligned} \Gamma_T^n &= p_{\text{rec},T} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} (1 - R_N(v', \vartheta')) \Gamma_{\nu-}^i(v', \vartheta', \varphi') dv' d\vartheta' d\varphi' \\ &\quad + p_{\text{refl},T} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} (1 - R_N(v', \vartheta')) \Gamma_{\nu-}^n(v', \vartheta', \varphi') dv' d\vartheta' d\varphi'. \end{aligned} \quad (3.84)$$

The net boundary fluxes for parallel momentum and energy are then identical as before, but with R^i replaced by $p_{\text{rec},F}^{\text{corr}}$, R^n replaced by $p_{\text{refl},F}^{\text{corr}}$, and the thermal particle flux Γ_T^n based on Eq. (3.84) instead of Eq. (3.47).

3.2.6 Practical implementation

After the theoretical derivation of the different boundary moment fluxes, we explain in this section how this is implemented in the finite-volume neutral code. Firstly, we explain how the different moment reflection coefficients R_N , $R_{m,||}$, $R_{m,\perp}$, R_p , R_{n_n} , $R_{u_{n\phi}}$ and R_E are extracted from the TRIM database. Subsequently, we evaluate the three-dimensional integrals over the velocity space to obtain the particle (Eqs. (3.44)-(3.47)), parallel momentum (Eqs. (3.50),(3.57),(3.62),(3.67)) and energy (Eq. (3.68)) fluxes.

TRIM moment reflection coefficients

We evaluate the moment reflection coefficients $R_\mu(v', \vartheta')$ for a finite number of values for v' and ϑ' , with the subscript μ indicating the particular moment. The velocity space is discretized logarithmically as

$$\ln v'_i = \ln \left(0.1 \sqrt{\frac{T_{a,\min}}{m}} \right) + \frac{i-1}{99} \left(\ln \left(10 \sqrt{\frac{T_{a,\max}}{m}} \right) - \ln \left(0.1 \sqrt{\frac{T_{a,\min}}{m}} \right) \right), \quad \text{for } i = 1, \dots, 100, \quad (3.85)$$

with $T_{a,\min}$ and $T_{a,\max}$ respectively the minimum and maximum occurring temperatures of the problem. The velocity is expressed in m/s. This discretization guarantees that all relevant particle energies in the plasma edge are taken into account. The polar angle is discretized uniformly as

$$\vartheta'_j = \frac{\pi}{400} + (j-1) \frac{\pi}{200}, \quad \text{for } j = 1, \dots, 100. \quad (3.86)$$

As explained in Section 3.1.2, the reflection coefficient is tabulated as a function of the incident energy and angle. To obtain $R_N(v'_i, \vartheta'_j)$, the speed is converted to the corresponding energy and the TRIM data is linearly inter- or extrapolated. Fig. 3.5 shows the interpolated particle reflection coefficient for carbon boundary material as a function of the incident energy and polar angle. The probability of reflection as a fast atom is the largest for particles with highly grazing incidence ($\vartheta' \approx 90^\circ$). The energy for which the particle reflection coefficient for a fixed incident angle peaks, also increases for increasing polar angle.

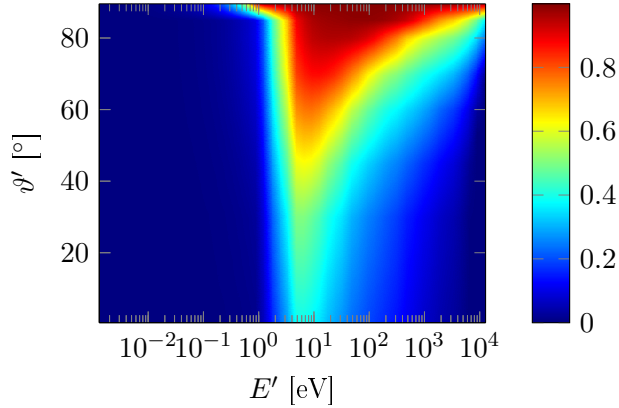


Figure 3.5: Particle reflection coefficient for deuterium projectiles on a carbon target.

It is harder to calculate the other moment reflection coefficients (elaborating the integrals of Eqs. (3.51),(3.52),(3.58),(3.63),(3.66) and (3.69)) due to the specific format of the TRIM tables. Therefore, we use a Monte Carlo simulation with a very large amount of particles. For every combination of v'_i and ϑ'_j , we interpolate or extrapolate the whole TRIM table (e.g., Fig. 3.3). Then, we launch a particle with speed v'_i , polar angle ϑ'_j and zero azimuthal angle and use the Monte Carlo procedure of Section 3.1.3. This leads to a reflected speed v_k , polar angle ϑ_k and azimuthal angle φ_k . The process is repeated for N particles. The moment reflection coefficient is estimated as

$$\hat{R}_\mu(v'_i, \vartheta'_j) = R_N(v'_i, \vartheta'_j) \frac{1}{\mu(v'_i, \vartheta'_j, 0)} \frac{1}{N} \sum_{k=1}^N \mu(v_k, \vartheta_k, \varphi_k), \quad (3.87)$$

with the hat on top of R indicating that it is a Monte Carlo estimation, and $\mu(v, \vartheta, \varphi)$ the particular moment, e.g., $\mu(v, \vartheta, \varphi) = v \sin \vartheta \cos \varphi$ for the parallel momentum reflection coefficient.

Fig. 3.6 shows the results for carbon boundary material. For every combination of v'_i and ϑ'_j , $N = 10^5$ particles are used. Most moment reflection coefficients behave similarly as the particle reflection coefficient (Fig. 3.5), with peak values for large incident angles and increase of the maximum value for fixed ϑ' when ϑ' increases. This is different for the density reflection coefficient (Fig. 3.6d) due to the $1/v$ scaling in Eq. (3.63).

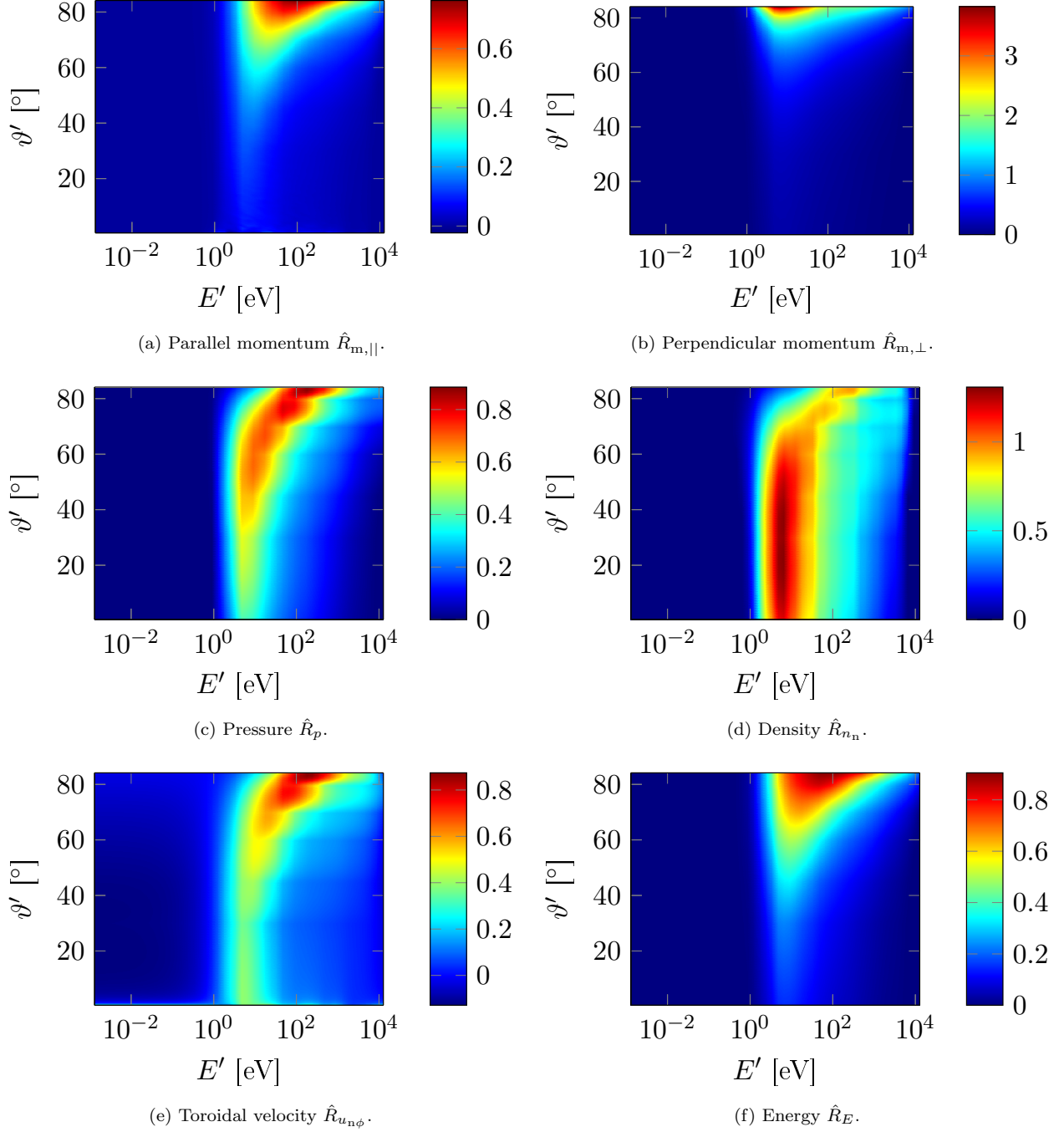


Figure 3.6: TRIM moment reflection coefficients for deuterium projectiles on a carbon target calculated with a Monte Carlo simulation.

Resulting fluid boundary fluxes

We have characterized the TRIM database with seven moment reflection coefficients. Now, the integrals of Section 3.2.3 have to be calculated.

For the contributions of the incident neutrals (subscript $\nu-$), this can be done analytically. This leads to following (particle, parallel momentum and energy flux densities):

$$\Gamma_{\nu-}^{\text{n,diff}} = - \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n M_{\nu}^i - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} M_{\tau\nu}^i + \frac{\partial n_n}{\partial \nu} M_{\nu\nu}^i \right) \right), \quad (3.88)$$

$$\begin{aligned} \Gamma_{\text{m},||,\nu-}^{\text{n,diff}} = & - m \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n (b_{\theta} (\sin \alpha_{\nu} M_{\tau\nu}^i + \cos \alpha_{\nu} M_{\nu\nu}^i) + b_{\phi} M_{\phi\nu}^i) \right. \\ & - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} (b_{\theta} (\sin \alpha_{\nu} M_{\tau\tau\nu}^i + \cos \alpha_{\nu} M_{\tau\nu\nu}^i) + b_{\phi} M_{\phi\tau\nu}^i) \right. \\ & \left. \left. + \frac{\partial n_n}{\partial \nu} (b_{\theta} (\sin \alpha_{\nu} M_{\nu\nu\tau}^i + \cos \alpha_{\nu} M_{\nu\nu\nu}^i) + b_{\phi} M_{\phi\nu\nu}^i) \right) \right), \end{aligned} \quad (3.89)$$

$$\begin{aligned} Q_{\nu-}^{\text{n,diff}} = & - \frac{m}{2} \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n (M_{\phi\phi\nu}^i + M_{\tau\tau\nu}^i + M_{\nu\nu\nu}^i) \right. \\ & - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} (M_{\phi\phi\tau\nu}^i + M_{\tau\tau\tau\nu}^i + M_{\tau\nu\nu\nu}^i) \right. \\ & \left. \left. + \frac{\partial n_n}{\partial \nu} (M_{\phi\phi\nu\nu}^i + M_{\tau\tau\nu\nu}^i + M_{\nu\nu\nu\nu}^i) \right) \right), \end{aligned} \quad (3.90)$$

with

$$M_{\alpha\beta\gamma\delta}^i = \int_{v'_{\phi}=-\infty}^{\infty} \int_{v'_{\tau}=-\infty}^{\infty} \int_{v'_{\nu}=-\infty}^0 v'_{\alpha} v'_{\beta} v'_{\gamma} v'_{\delta} \widetilde{M}(v'_{\phi}, v'_{\tau}, v'_{\nu}, \mathbf{V}_i, T_i) dv'_{\phi} dv'_{\tau} dv'_{\nu}, \quad (3.91)$$

i.e., a certain moment of a Maxwellian distribution, with the subscripts indicating the particular velocity components by which the Maxwellian is multiplied. When using the drifting Maxwellian assumption instead of the diffusion assumption we get

$$\Gamma_{\nu-}^{\text{n,maxw}} = - n_n M_{\text{n},\nu}^{\text{n}}, \quad (3.92)$$

$$\Gamma_{\text{m},||,\nu-}^{\text{n,maxw}} = - m \left(n_n (b_{\theta} (\sin \alpha_{\nu} M_{\tau\nu}^{\text{n}} + \cos \alpha_{\nu} M_{\nu\nu}^{\text{n}}) + b_{\phi} M_{\phi\nu}^{\text{n}}) \right), \quad (3.93)$$

$$Q_{\nu-}^{\text{n,maxw}} = - \frac{m}{2} \left(n_n (M_{\phi\phi\nu}^{\text{n}} + M_{\tau\tau\nu}^{\text{n}} + M_{\nu\nu\nu}^{\text{n}}) \right), \quad (3.94)$$

with

$$M_{\alpha\beta\gamma\delta}^{\text{n}} = \int_{v'_{\phi}=-\infty}^{\infty} \int_{v'_{\tau}=-\infty}^{\infty} \int_{v'_{\nu}=-\infty}^0 v'_{\alpha} v'_{\beta} v'_{\gamma} v'_{\delta} \widetilde{M}(v'_{\phi}, v'_{\tau}, v'_{\nu}, \mathbf{V}_n, T_n) dv'_{\phi} dv'_{\tau} dv'_{\nu} \quad (3.95)$$

To obtain the fluid parallel momentum flux density of the incident neutrals $\Gamma_{\text{m},\text{f},||,\nu-}$, we have to subtract the pressure contribution from Eq. (3.89), as expressed by Eq. (3.54). Therefore, we also have to calculate the contributions from the incident neutrals to $\frac{m}{3} \int ||\mathbf{v}'||^2 \frac{\Gamma_{\text{n}}^{\text{n}}(\mathbf{v}')}{|v'_{\nu}|} d\mathbf{v}'$ (Eq. (3.57)), n'_{n} (Eq. (3.62)) and $n'_{\text{n}} u'_{\text{n}\phi}$

(Eq. (3.67)). Elaborating these terms, gives for the diffusion approach

$$\left[\frac{m}{3} \int \frac{||\mathbf{v}||^2}{|v_\nu|} \Gamma^n(\mathbf{v}) d\mathbf{v} \right]_{\nu-}^{\text{diff}} = \frac{m}{3} \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n (M_{\phi\phi}^i + M_{\tau\tau}^i + M_{\nu\nu}^i) \right. \\ \left. - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} (M_{\phi\phi\tau}^i + M_{\tau\tau\tau}^i + M_{\nu\nu\tau}^i) \right. \right. \\ \left. \left. + \frac{\partial n_n}{\partial \nu} (M_{\phi\phi\nu}^i + M_{\tau\tau\nu}^i + M_{\nu\nu\nu}^i) \right) \right), \quad (3.96)$$

$$(n'_{n,\nu-})^{\text{diff}} = \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n M^i \right. \\ \left. - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} M_\tau^i + \frac{\partial n_n}{\partial \nu} M_\nu^i \right) \right), \quad (3.97)$$

$$[n'_n u'_{n\phi}]_{\nu-}^{\text{diff}} = \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n M_\phi^i \right. \\ \left. - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} M_{\phi\tau}^i + \frac{\partial n_n}{\partial \nu} M_{\phi\nu}^i \right) \right). \quad (3.98)$$

For the drifting Maxwellian approach this becomes

$$\left[\frac{m}{3} \int \frac{||\mathbf{v}||^2}{|v_\nu|} \Gamma^n(\mathbf{v}) d\mathbf{v} \right]_{\nu-}^{\text{maxw}} = \frac{m}{3} (n_n (M_{\phi\phi}^n + M_{\tau\tau}^n + M_{\nu\nu}^n)), \quad (3.99)$$

$$(n'_{n,\nu-})^{\text{maxw}} = n_n M^n \quad (3.100)$$

$$[n'_n u'_{n\phi}]_{\nu-}^{\text{maxw}} = n_n M_\phi^n. \quad (3.101)$$

For the fast recycled and reflected contributions, the integrals can no longer be calculated analytically and they are evaluated numerically. To avoid the cost of recalculating the integrals every time the plasma state changes, the integrals are evaluated in advance for different values of the influencing plasma parameters. Every boundary flux density Γ_μ^n of a particular moment μ can be written as

$$\Gamma_\mu^n = \Gamma_{\mu,\nu-}^n + \sum_k A_\mu^k \int_{v'=0}^\infty \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} g_\mu^k(v', \vartheta', \varphi'; \mathbf{R}) \dots \\ \dots \tilde{f}_a(v', \vartheta', \varphi'; \delta_{\text{sh}}^{\text{pot},k}, \mathbf{V}_a, T_a) dv' d\vartheta' d\varphi', \quad (3.102)$$

with $\Gamma_{\mu,\nu-}^n$ the analytic expressions for the incident contributions (Eqs. (3.88)-(3.90),(3.96)-(3.98)). The coefficient A_μ^k contains all terms that can be brought outside the integral (no function of v' , ϑ' or φ'). The summation is used to distinguish between the different contributions. There is a contribution due to the recycled neutrals and three contributions of the reflected neutrals, respectively scaling with n_n , $\frac{\partial n_n}{\partial \tau}$ and $\frac{\partial n_n}{\partial \nu}$ (see Eq. (3.39)). The integrand can be written as the product of a function $g_\mu^k(v', \vartheta', \varphi'; \mathbf{R})$, which contains the needed TRIM reflection coefficients (present in $\mathbf{R}$). For the recycling contributions and the reflection contributions using the diffusion approximation, the Maxwellian distribution $\tilde{f}_a(v', \vartheta', \varphi'; \delta_{\text{sh}}^{\text{pot},k}, \mathbf{V}_a, T_a) dv' d\vartheta' d\varphi'$ is equal to ion distribution $\tilde{f}_i(v', \vartheta', \varphi'; \delta_{\text{sh}}^{\text{pot},k}, \mathbf{V}_i, T_i)$ given by Eq. (3.12). The superscript k for $\delta_{\text{sh}}^{\text{pot}}$ is needed to distinguish between the recycled and reflected neutrals. For the reflected neutrals using the diffusion approximation, $\delta_{\text{sh}}^{\text{pot},k} = 0$, whereas $\delta_{\text{sh}}^{\text{pot},k}$ is nonzero for the recycled neutrals if the boundary is not fully parallel to the magnetic field. For the reflected neutrals with the drifting Maxwellian assumption, $\tilde{f}_a(v', \vartheta', \varphi'; \delta_{\text{sh}}^{\text{pot},k}, \mathbf{V}_a, T_a) dv' d\vartheta' d\varphi'$ is equal to $\tilde{f}_n(v', \vartheta', \varphi'; 0, \mathbf{V}_n, T_n) dv' d\vartheta' d\varphi'$, given by Equation (3.42). Only the integral in Eq. (3.102) is calculated in advance and saved as a function of the sheath transmission

coefficient, ion fluid velocity and temperature:

$$\begin{aligned} \text{RC}_\mu^k(\mathbf{V}_a, T_a) &= \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} g_\mu^k(v', \vartheta', \varphi'; \mathbf{R}) \dots \\ &\dots \tilde{f}_a(v', \vartheta', \varphi'; \delta_{\text{sh}}^{\text{pot}, k}, \mathbf{V}_a, T_a) dv' d\vartheta' d\varphi'. \end{aligned} \quad (3.103)$$

To reduce the number of dependencies of RC_μ^k , we assume that $\mathbf{V}_a \approx [u_{a,\phi} \ 0 \ 0]^T$, expressed in the (ϕ, τ, ν) coordinate system. This assumption is valid for the ions if the pitch is sufficiently small such that the parallel direction almost coincides with the toroidal direction, and drifts are not included. For the neutrals in the drifting Maxwellian approximation, or for the ions when drifts are included, this assumption is less accurate. Even so, good fluid-kinetic agreement was achieved for cases with drifts in Ref. [27]. Studying the role of the perpendicular bulk velocities, and finding a way to include them without making the databases unacceptably large, is subject of future work. We evaluate Eq. (3.103) for 100 values of the Mach number $\mathcal{M}_{a,\phi} = u_{a,\phi}/\sqrt{2T_a/m}$, uniformly distributed between 0 and 1.1 and 100 values for the temperature logarithmically distributed between 0.1 and 1000 eV. The midpoint integration rule is used with the discretized speed and polar angle from respectively Eq. (3.85) and Eq. (3.86), and the azimuthal angle φ' discretized as

$$\varphi'_l = \frac{\pi}{400} + (l-1)\frac{\pi}{200}, \quad \text{for } l = 1, \dots, 400. \quad (3.104)$$

Now, all integrated reflection coefficients are tabulated as a function of $\mathcal{M}_{a,\phi}$, T_a , and $\delta_{\text{sh}}^{\text{pot}}$. During the simulation, we use linear interpolation for intermediate values of $\mathcal{M}_{a,\phi}$, T_a , and $\delta_{\text{sh}}^{\text{pot}}$.

Chapter 4

Implementation in SOLPS-ITER

4.1 Final expressions for macroscopic AFN database coefficients

The final expressions for the pre-computed integrals, as calculated in `CalcReflectionCoefficientsFluidTRIM`, are given here. In general, there are four coefficients per TRIM reflection coefficient. There is a contribution due to the recycled neutrals and three contributions of the reflected neutrals, respectively scaling with n_n , $\frac{\partial n_n}{\partial \tau}$ and $\frac{\partial n_n}{\partial \nu}$ (see Eq. (3.39)). However, the reflection term scaling with n_n leads to the same integral as the recycling term scaling with n_i . These coefficients are called RC_n , $RC_{m,\phi}$, $RC_{m,\nu}$, RC_p , RC_{n_n} , $RC_{u_n,\phi}$, and RC_E . The terms scaling with $\frac{\partial n_n}{\partial \tau}$ are often zero because $\int_{\varphi'=0}^{2\pi} A \sin(\varphi') \exp(B + C \cos(\varphi')) d\varphi' = 0$. The exception is the tangential momentum components, where both the terms scaling with n_n and $\frac{\partial n_n}{\partial \nu}$ are zero and only the $\frac{\partial n_n}{\partial \tau}$ term remains. The non-zero coefficients scaling with a neutral density gradient are $RC_n^{\frac{\partial n_n}{\partial \nu}}$, $RC_{m,\phi}^{\frac{\partial n_n}{\partial \nu}}$, $RC_{m,\tau}^{\frac{\partial n_n}{\partial \nu}}$, $RC_{m,\nu}^{\frac{\partial n_n}{\partial \nu}}$, $RC_p^{\frac{\partial n_n}{\partial \nu}}$, $RC_{n_n}^{\frac{\partial n_n}{\partial \nu}}$, $RC_{u_n,\phi}^{\frac{\partial n_n}{\partial \nu}}$, and $RC_E^{\frac{\partial n_n}{\partial \nu}}$. The coefficients which are zero and hence not tabulated are $RC_n^{\frac{\partial n_n}{\partial \tau}}$, $RC_{m,\phi}^{\frac{\partial n_n}{\partial \tau}}$, $RC_{m,\nu}^{\frac{\partial n_n}{\partial \tau}}$, $RC_{m,\tau}^{\frac{\partial n_n}{\partial \tau}}$, $RC_{m,\nu}^{\frac{\partial n_n}{\partial \tau}}$, $RC_p^{\frac{\partial n_n}{\partial \tau}}$, $RC_{n_n}^{\frac{\partial n_n}{\partial \tau}}$, $RC_{u_n,\phi}^{\frac{\partial n_n}{\partial \tau}}$, and $RC_E^{\frac{\partial n_n}{\partial \tau}}$.

The coefficients below are written as a function of $u_{a,\phi}$ and T_a . When evaluating the coefficients in `CalcRecycledFluxes` and `CalcReflectedFluxesDiffusion`, $a = i$. When evaluating the coefficients in `CalcReflectedFluxesMaxwellian`, $a = n$. Only for the recycling contributions δ_{sh}^{pot} is nonzero.

The term $\left(\frac{m}{2\pi T_a}\right)^{\frac{3}{2}}$ in front of the exponential in the (sheath-accelerated) Maxwellian is rewritten as $\pi^{-\frac{3}{2}} c_a^{-3}$, where c_a is the isothermal sound speed of species a . The coefficients are further divided by powers of c_a to make them dimensionless, i.e. of the form

$$RC = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^k} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R(v')^{k-1} F(\vartheta') G(\varphi') dv' d\vartheta' d\varphi'. \quad (4.1)$$

The sheath acceleration is determined by $\Delta\phi_{sh} = \delta_{sh}^{\text{pot}} T_e e^{-1}$, while the macroscopic reflection coefficients are tabulated as a function of $T_{i/n}$. This is solved by evaluating the database for

$$\bar{\delta}_{sh}^{\text{pot}} = \delta_{sh}^{\text{pot}} \frac{T_e}{T_{i/n}} = \frac{\Delta\phi_{sh} e}{T_{i/n}}. \quad (4.2)$$

A final point is that the reflection coefficients scaling with $\frac{\partial n_n}{\partial \nu}$ or $\frac{\partial n_n}{\partial \tau}$ will always be evaluated for $\delta_{sh}^{\text{pot}} = \bar{\delta}_{sh}^{\text{pot}} = 0$. However, they are currently also tabulated as a function of δ_{sh}^{pot} such that all coefficients have the same dimension.

4.1.1 Macroscopic particle reflection coefficients

Equations (3.45), (3.46), (3.33), and (3.39) lead to the definition of the macroscopic particle reflection coefficient scaling with density

$$\begin{aligned} \text{RC}_n(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^4} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta')(v')^3 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta') dv' d\vartheta' d\varphi', \end{aligned} \quad (4.3)$$

and the macroscopic particle reflection coefficient scaling with the normal gradient of the neutral density

$$\begin{aligned} \text{RC}_n^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta')^2 \sin(\vartheta') dv' d\vartheta' d\varphi'. \end{aligned} \quad (4.4)$$

The macroscopic particle reflection coefficient scaling with the tangential gradient of the neutral density is zero and hence not tabulated:

$$\begin{aligned} \text{RC}_n^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_N(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^2 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \end{aligned} \quad (4.5)$$

4.1.2 Macroscopic momentum reflection coefficients

Equations (3.33), (3.39), and the term in Eq. (3.50) that scales with b_ϕ lead to the definition of the macroscopic toroidal momentum reflection coefficient scaling with density:

$$\begin{aligned} \text{RC}_{m,\phi}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,||}(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^2 \cos(\varphi') dv' d\vartheta' d\varphi', \end{aligned} \quad (4.6)$$

and the macroscopic toroidal momentum reflection coefficient scaling with the normal gradient of neutral density

$$\begin{aligned} \text{RC}_{m,\phi}^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,||}(v', \vartheta')(v')^5 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) (\cos(\vartheta') \sin(\vartheta'))^2 \cos(\varphi') dv' d\vartheta' d\varphi'. \end{aligned} \quad (4.7)$$

The macroscopic toroidal momentum reflection coefficient scaling with the tangential gradient of neutral density is zero and hence not tabulated:

$$\begin{aligned} \text{RC}_{m,\phi}^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,||}(v', \vartheta')(v')^5 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^3 \cos(\varphi') \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \end{aligned} \quad (4.8)$$

Equations (3.33), (3.39), and the term in Eq. (3.50) that scales with $b_\theta \sin \alpha_\nu$ lead to the definition of the macroscopic tangential momentum reflection coefficients. Now, the part scaling with density is zero:

$$\begin{aligned} R_{m,\tau}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,||}(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^2 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \end{aligned} \quad (4.9)$$

Also the part scaling with the normal gradient of neutral density is zero:

$$\begin{aligned} R_{m,\tau}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,||}(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^2 \sin(\varphi') \cos(\varphi') dv' d\vartheta' d\varphi' = 0. \end{aligned} \quad (4.10)$$

The macroscopic tangential momentum reflection coefficients scaling with the tangential gradient of neutral density is

$$\begin{aligned} \text{RC}_{m,\tau}^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,||}(v', \vartheta')(v')^5 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^3 \sin(\varphi')^2 dv' d\vartheta' d\varphi'. \end{aligned} \quad (4.11)$$

Equations (3.33), (3.39), and the term in Eq. (3.50) that scales with $b_\theta \cos \alpha_\nu$ lead to the definition of the macroscopic normal momentum reflection coefficient scaling with density

$$\begin{aligned} \text{RC}_{m,\nu}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,\perp}(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta')^2 \sin(\vartheta') dv' d\vartheta' d\varphi', \end{aligned} \quad (4.12)$$

and the macroscopic normal momentum reflection coefficient scaling with the normal gradient of neutral density

$$\begin{aligned} \text{RC}_{m,\nu}^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{m,\perp}(v', \vartheta')(v')^5 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta')^3 \sin(\vartheta') dv' d\vartheta' d\varphi'. \end{aligned} \quad (4.13)$$

The macroscopic normal momentum reflection coefficient scaling with the tangential gradient of neutral density is zero and hence not tabulated:

$$\begin{aligned} \text{RC}_{\text{m},\nu}^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{\text{m},\perp}(v', \vartheta')(v')^5 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta')^2 \sin(\vartheta')^2 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \end{aligned} \quad (4.14)$$

4.1.3 Pressure contributions

Equations (3.54), (3.55) and (3.56) introduce the need for additional coefficients to calculate the pressure of the fast reflected and recycled neutrals.

Equations (3.57), (3.39) and (3.33) lead to the definition of the part of the macroscopic pressure reflection coefficient scaling with density

$$\begin{aligned} \text{RC}_p(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_p(v', \vartheta')(v')^4 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \sin(\vartheta') dv' d\vartheta' d\varphi', \end{aligned} \quad (4.15)$$

and the part of the macroscopic pressure reflection coefficient scaling with the normal gradient of the neutral density

$$\begin{aligned} \text{RC}_p^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_p(v', \vartheta')(v')^5 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta') dv' d\vartheta' d\varphi'. \end{aligned} \quad (4.16)$$

The part of the macroscopic pressure reflection coefficient scaling with the tangential gradient of the neutral density is again zero:

$$\begin{aligned} \text{RC}_p^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^3} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_p(v', \vartheta')(v')^2 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \sin(\vartheta')^2 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \end{aligned} \quad (4.17)$$

Equations (3.62), (3.39) and (3.33) lead to the definition of the part of the macroscopic density reflection coefficient scaling with density

$$\begin{aligned} \text{RC}_{n_n}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) &= \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^4} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{n_n}(v', \vartheta')(v')^3 \\ &\exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \sin(\vartheta') dv' d\vartheta' d\varphi', \end{aligned} \quad (4.18)$$

and the part of the macroscopic density reflection coefficient scaling with the normal gradient of the neutral density:

$$\text{RC}_{n_n}^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^4} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{n_n}(v', \vartheta')(v')^3 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta') dv' d\vartheta' d\varphi'. \quad (4.19)$$

The part of the macroscopic density reflection coefficient scaling with the tangential gradient of the neutral density is again zero:

$$\text{RC}_{n_n}^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^4} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{n_n}(v', \vartheta')(v')^3 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \sin(\vartheta')^2 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \quad (4.20)$$

Equations (3.67), (3.39) and (3.33) lead to the definition of part of the macroscopic toroidal velocity reflection coefficient scaling with density

$$\text{RC}_{u_{n,\phi}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^4} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{u_{n,\phi}}(v', \vartheta')(v')^3 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \sin(\vartheta')^2 \cos(\varphi') dv' d\vartheta' d\varphi', \quad (4.21)$$

and the part of the macroscopic toroidal velocity reflection coefficient scaling with the normal gradient of neutral density:

$$\text{RC}_{u_{n,\phi}}^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{u_{n,\phi}}(v', \vartheta')(v')^4 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^2 \cos(\varphi') dv' d\vartheta' d\varphi'. \quad (4.22)$$

The part of the macroscopic toroidal velocity reflection coefficient scaling with the tangential gradient of neutral density is again zero:

$$\text{RC}_{u_{n,\phi}}^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^5} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_{u_{n,\phi}}(v', \vartheta')(v')^4 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^3 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \quad (4.23)$$

4.1.4 Macroscopic neutral energy reflection coefficients

Equations (3.68), (3.39) and (3.33) lead to the definition of the macroscopic energy reflection coefficient scaling with density

$$\text{RC}_E(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^6} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_E(v', \vartheta')(v')^5 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta')^2 \sin(\vartheta') dv' d\vartheta' d\varphi', \quad (4.24)$$

and the macroscopic energy reflection coefficient scaling with the normal gradient of the neutral density

$$\text{RC}_E^{\frac{\partial n_n}{\partial \nu}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^7} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_E(v', \vartheta')(v')^6 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta')^2 \sin(\vartheta') dv' d\vartheta' d\varphi'. \quad (4.25)$$

The macroscopic energy reflection coefficient scaling with the tangential gradient of the neutral density is zero:

$$\text{RC}_E^{\frac{\partial n_n}{\partial \tau}}(u_{a,\phi}, T_a, \delta_{\text{sh}}^{\text{pot}}) = \frac{1}{\pi^{\frac{3}{2}}} \frac{1}{c_a^7} \int_{v'=0}^{\infty} \int_{\vartheta'=0}^{\pi/2} \int_{\varphi'=0}^{2\pi} R_E(v', \vartheta')(v')^6 \exp\left(-\frac{m}{2T_a}((v')^2 - 2(v')u_{a,\phi}\sin(\vartheta')\cos(\varphi') + u_{a,\phi}^2 - \frac{2\delta_{\text{sh}}^{\text{pot}}T_a}{m})\right) \cos(\vartheta') \sin(\vartheta')^2 \sin(\varphi') dv' d\vartheta' d\varphi' = 0. \quad (4.26)$$

4.2 Final expressions for net boundary fluxes

Here, we re-write the expressions from the previous chapter in a form that closest resembles how they are implemented in the code. Recurring steps are:

- The moments of the thermally released part $\Gamma_{\mu,T}^{\text{n,diff/maxw}}$ is split into $\Gamma_{\mu,T,\text{refl}}^{\text{n,diff/maxw}} + \Gamma_{\mu,T,\text{rec}}^{\text{n,diff/maxw}}$, where $\Gamma_{\mu,T,\text{refl}}^{\text{n,diff/maxw}}$ corresponds to the thermally released atoms originating from incident atoms, and $\Gamma_{\mu,T,\text{rec}}^{\text{n,diff/maxw}}$ corresponds to the thermally released atoms originating from incident ions.
- The recycled parts are multiplied with $\frac{\Gamma_{\nu-}^i}{-M_\nu}$. This is related to the factor C^i introduced in Equation (3.6). It compensates for the fact that integrating the assumed ion velocity distribution (that was used to determine the macroscopic fast recycling coefficients) can lead to a slightly different result than the macroscopic target ion flux in the fluid plasma solver.
- R^i and R^n are replaced by $p_{\text{rec},F}^{\text{corr}}$ and $p_{\text{refl},F}^{\text{corr}}$, respectively, as explained in Section 3.2.5.

4.2.1 Net neutral particle flux density

The net neutral particle flux density $\Gamma^n = n_n \mathbf{V}_n \cdot \boldsymbol{\nu}$ is written as

$$\Gamma^n = \Gamma_{\text{refl},F}^{\text{n,diff/maxw}} + \Gamma_{\text{rec},F}^{\text{n}} + \Gamma_{\text{T},\text{refl}}^{\text{n,diff/maxw}} + \Gamma_{\text{T},\text{rec}}^{\text{n,diff/maxw}} - \Gamma_{\nu-}^{\text{n,maxw/diff}}. \quad (4.27)$$

Here, $\Gamma_{\nu-}^{\text{n,diff}}$ and $\Gamma_{\nu-}^{\text{n,maxw}}$ were already given as Eqs. (3.88) and (3.92). They are implemented in `CalcIncidentFluxesDiffusion` and `CalcIncidentFluxesMaxwellian`, respectively. For the recycled flux we have

$$\Gamma_{\text{rec},f} + \Gamma_{\text{T},\text{rec}}^{\text{n}} = R_{\text{fl}}^i(\mathbf{V}_i \cdot \boldsymbol{\nu}), \quad (4.28)$$

as calculated in `CalcRecycledFluxes`. For the reflected fluxes we have

$$\Gamma_{\text{refl},f}^{\text{n,diff}} + \Gamma_{\text{T}}^{\text{n,diff}} = R_{\text{fl}}^{\text{n}} \Gamma_{\nu-}^{\text{n,diff}} \quad (4.29)$$

as implemented `CalcReflectedFluxesDiffusion` or

$$\Gamma_{\text{refl},f}^{\text{n,maxw}} + \Gamma_{\text{T}}^{\text{n,maxw}} = R_{\text{fl}}^{\text{n}} \Gamma_{\nu-}^{\text{n,maxw}} \quad (4.30)$$

as implemented in `CalcReflectedFluxesMaxwellian`.

4.2.2 Net neutral parallel momentum flux density

The net neutral parallel momentum flux given by Equations (3.54), (3.50), and (3.53) is finally implemented as

$$\begin{aligned} \Gamma_{m,||,f}^{n,diff/maxw} = & \Gamma_{m,||,rec}^n + \Gamma_{m,||,refl}^{n,diff/maxw} + \Gamma_{m,||,T,rec}^{n,diff/maxw} + \Gamma_{m,||,T,refl}^{n,diff/maxw} - \Gamma_{m,||,inc}^{n,diff/maxw} \\ & - b_\theta \cos \alpha_\nu (p_{1,rec}^{n,diff/maxw} + p_{1,refl}^{n,diff/maxw} + p_{1,T,rec}^n + p_{1,T,refl}^{n,diff/maxw} - p_{1,T,inc}^{n,diff/maxw} + p_2) \end{aligned} \quad (4.31)$$

The p_1 terms correspond to term (1) in Equation (3.56). The p_2 term correspond to term (2) in Equation (3.56) and is further elaborated below.

Flux contributions

Using the definition of $RC_{m,\nu}$ (Eq. (4.12)) and $RC_{m,\phi}$ (Eq. (4.6)), the fast recycled part of Eq. (3.50) is written as

$$\Gamma_{m,||,rec}^n = b_\theta (\Gamma_{m,\nu+,rec}^n \cos \alpha_\nu - \Gamma_{m,\tau,rec}^n \sin \alpha_\nu) - b_\phi \Gamma_{m,\phi,rec}^n \quad (4.32)$$

with

$$\Gamma_{m,\nu+,rec}^n = p_{rec,F}^{corr} m RC_{m,\nu} c_i^2 \frac{\Gamma_{\nu-}^i}{-M_\nu}, \quad (4.33)$$

$$\Gamma_{m,\tau,rec}^n = 0, \quad (4.34)$$

$$\Gamma_{m,\phi,rec}^n = p_{rec,F}^{corr} m RC_{m,\phi} c_i^2 \frac{\Gamma_{\nu-}^i}{-M_\nu}, \quad (4.35)$$

where $p_{rec,F}^{corr} = \frac{\min(p_{rec,F}, R_{fl}^i)}{p_{rec,F}}$ with

$$p_{rec,F} = \frac{RC_n c_i}{-M_\nu}. \quad (4.36)$$

Using the definition of $RC_{m,\nu}$ (Eq. (4.12)), $RC_{m,\nu}^{\frac{\partial n_n}{\partial \nu}}$ (Eq. (4.13)), $RC_{m,\tau}^{\frac{\partial n_n}{\partial \tau}}$ (Eq. (4.11)), $RC_{m,\phi}$ (Eq. (4.6)), and $RC_{m,\phi}^{\frac{\partial n_n}{\partial \nu}}$ (Eq. (4.7)), the fast reflected part of Eq. (3.50) according to the diffusion approximation is written as

$$\Gamma_{m,||,refl}^{n,diff} = b_\theta (\Gamma_{m,\nu+,refl}^{n,diff} \cos \alpha_\nu - \Gamma_{m,\tau,refl}^{n,diff} \sin \alpha_\nu) - b_\phi \Gamma_{m,\phi,refl}^{n,diff} \quad (4.37)$$

with

$$\Gamma_{m,\nu+,refl}^{n,diff} = p_{refl,F}^{corr,diff} m \frac{n_i K_{cx,m}}{n_i K_{cx,m} + n_e K_i} (n_n RC_{m,\nu} c_i^2 + \frac{1}{n_i K_{cx,m} + n_e K_i} \frac{\partial n_n}{\partial \nu} RC_{m,\nu}^{\frac{\partial n_n}{\partial \nu}} c_i^3), \quad (4.38)$$

$$\Gamma_{m,\tau,refl}^{n,diff} = p_{refl,F}^{corr,diff} m \frac{n_i K_{cx,m}}{(n_i K_{cx,m} + n_e K_i)^2} \frac{\partial n_n}{\partial \tau} RC_{m,\tau}^{\frac{\partial n_n}{\partial \tau}} c_i^3, \quad (4.39)$$

$$\Gamma_{m,\phi,refl}^{n,diff} = p_{refl,F}^{corr,diff} m \frac{n_i K_{cx,m}}{n_i K_{cx,m} + n_e K_i} (n_n RC_{m,\phi} c_i^2 + \frac{1}{n_i K_{cx,m} + n_e K_i} \frac{\partial n_n}{\partial \nu} RC_{m,\phi}^{\frac{\partial n_n}{\partial \nu}} c_i^3), \quad (4.40)$$

where $p_{refl,F}^{corr,diff} = \frac{\min(p_{refl,F}^{diff}, R_{fl}^n)}{p_{refl,F}^{diff}}$ with

$$p_{refl,F}^{diff} = \frac{n_i K_{cx,m}}{n_i K_{cx,m} + n_e K_i} (n_n RC_n c_i + \frac{1}{n_i K_{cx,m} + n_e K_i} \frac{\partial n_n}{\partial \nu} RC_n^{\frac{\partial n_n}{\partial \nu}} c_i^2) \frac{1}{\Gamma_{\nu-}^{n,diff}}. \quad (4.41)$$

When using the drifting Maxwellian approximation, the fast reflected parallel momentum flux becomes

$$\Gamma_{m,||,refl}^{n,maxw} = b_\theta (\Gamma_{m,\nu+,refl}^{n,maxw} \cos \alpha_\nu - \Gamma_{m,\tau,refl}^{n,maxw} \sin \alpha_\nu) - b_\phi \Gamma_{m,\phi,refl}^{n,maxw} \quad (4.42)$$

with

$$\Gamma_{m,\nu+,refl}^{n,maxw} = p_{refl,F}^{corr,maxw} m n_n RC_{m,\nu} c_n^2, \quad (4.43)$$

$$\Gamma_{m,\tau,refl}^{n,maxw} = 0, \quad (4.44)$$

$$\Gamma_{m,\phi,refl}^{n,maxw} = p_{refl,F}^{corr,maxw} m n_n RC_{m,\phi} c_n^2. \quad (4.45)$$

where $p_{refl,F}^{corr,maxw} = \frac{\min(p_{refl,F}^{maxw}, R_n^n)}{p_{refl,F}^{maxw}}$ with

$$p_{refl,F}^{maxw} = \frac{n_n RC_n c_n}{\Gamma_{\nu-}^{n,maxw}}. \quad (4.46)$$

The thermally recycled contribution to the parallel momentum flux from Eq. (3.53) is written as

$$\Gamma_{m,||,T,rec}^n = b_\theta \Gamma_{m,\nu+,T,rec}^n \cos \alpha_\nu \quad (4.47)$$

with

$$\Gamma_{m,\nu+,T,rec}^n = p_{rec,T} \frac{2}{3} m \Gamma_{\nu-}^i v_T, \quad (4.48)$$

where $p_{rec,T} = \max(R_{fl}^i - p_{rec,F}, 0)$.

The thermally reflected contribution to the parallel momentum flux from Eq. (3.53) is written as

$$\Gamma_{m,||,T,refl}^{n,diff} = b_\theta \Gamma_{m,\nu+,T,refl}^{n,diff} \cos \alpha_\nu \quad (4.49)$$

with

$$\Gamma_{m,\nu+,T,refl}^{n,diff} = p_{refl,T}^{diff} \frac{2}{3} m \Gamma_{\nu-}^{n,diff} v_T \quad (4.50)$$

where $p_{refl,T}^{diff} = \max(R_{fl}^n - p_{refl,F}^{diff}, 0)$. When using the drifting Maxwellian assumption, this becomes

$$\Gamma_{m,||,T,refl}^{n,maxw} = b_\theta \Gamma_{m,\nu+,T,refl}^{n,maxw} \cos \alpha_\nu \quad (4.51)$$

with

$$\Gamma_{m,\nu+,T,refl}^{n,maxw} = p_{refl,T}^{maxw} \frac{2}{3} m \Gamma_{\nu-}^{n,maxw} v_T. \quad (4.52)$$

The incident flux of parallel momentum according to the diffusion approximation (Eq. (3.89)) is written as

$$\Gamma_{m,||,inc}^{n,diff} = b_\theta (\Gamma_{m,\nu-,inc}^{n,diff} \cos \alpha_\nu - \Gamma_{m,\tau,inc}^{n,diff} \sin \alpha_\nu) - b_\phi \Gamma_{m,\phi,inc}^{n,diff} \quad (4.53)$$

with

$$\Gamma_{m,\nu-,inc}^{n,diff} = m \frac{n_i K_{cx,m}}{n_i K_{cx,m} + n_e K_i} \left(-n_n M_{\nu\nu} + \frac{1}{n_i K_{cx,m} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} u_\tau M_{\nu\nu} + \frac{\partial n_n}{\partial \nu} M_{\nu\nu\nu} \right) \right), \quad (4.54)$$

$$\Gamma_{m,\tau,inc}^{n,diff} = m \frac{n_i K_{cx,m}}{n_i K_{cx,m} + n_e K_i} \left(-n_n u_\tau M_\nu + \frac{1}{n_i K_{cx,m} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} \left(\frac{T_i}{m} + u_\tau^2 \right) M_\nu + \frac{\partial n_n}{\partial \nu} u_\tau M_{\nu\nu} \right) \right), \quad (4.55)$$

$$\Gamma_{m,\phi,inc}^{n,diff} = m \frac{n_i K_{cx,m}}{n_i K_{cx,m} + n_e K_i} \left(-n_n u_\phi M_\nu + \frac{1}{n_i K_{cx,m} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} u_\phi u_\tau M_\nu + \frac{\partial n_n}{\partial \nu} u_\phi M_{\nu\nu} \right) \right). \quad (4.56)$$

When using the drifting Maxwellian approximation, this becomes

$$\Gamma_{m,||,inc}^{n,maxw} = -b_\theta (\Gamma_{m,\nu-,inc}^{n,maxw} \cos \alpha_\nu + \Gamma_{m,\tau,inc}^{n,maxw} \sin \alpha_\nu) - b_\phi \Gamma_{m,\phi,inc}^{n,maxw} \quad (4.57)$$

with

$$\Gamma_{m,\nu-,inc}^{n,maxw} = -m n_n M_{\nu\nu}, \quad (4.58)$$

$$\Gamma_{m,\tau,inc}^{n,maxw} = -m n_n u_{n\tau} M_\nu, \quad (4.59)$$

$$\Gamma_{m,\phi,inc}^{n,maxw} = -m n_n u_{n\phi} M_\nu. \quad (4.60)$$

Term (1) pressure contributions

Using the definition of RC_p (Eq. (4.15)) and $\text{RC}_p^{\frac{\partial n_n}{\partial \nu}}$ (Eq. (4.16)), the term (1) pressure contributions from equation (3.57) as written in Eq. (4.31) become

$$p_{1,\text{rec}}^n = p_{\text{rec},\text{F}}^{\text{corr}} \frac{m}{3} \text{RC}_p c_i^2 \frac{\Gamma_{\nu-}^i}{-M_\nu}, \quad (4.61)$$

$$p_{1,\text{refl}}^{\text{n,diff}} = p_{\text{refl},\text{F}}^{\text{corr,diff}} \frac{m}{3} \frac{n_i K_{\text{cx},\text{m}}}{n_i K_{\text{cx},\text{m}} + n_e K_i} \left(n_n + \text{RC}_p c_i^2 + \frac{1}{n_i K_{\text{cx},\text{m}} + n_e K_i} \frac{\partial n_n}{\partial \nu} \text{RC}_p^{\frac{\partial n_n}{\partial \nu}} c_i^3 \right), \quad (4.62)$$

$$p_{1,\text{refl}}^{\text{n,maxw}} = p_{\text{refl},\text{F}}^{\text{corr,maxw}} \frac{m}{3} n_n \text{RC}_p c_n^2, \quad (4.63)$$

$$p_{1,\text{T},\text{rec}}^n = p_{\text{rec},\text{T}} \frac{2}{3} m \Gamma_{\nu-}^i v_{\text{T}}, \quad (4.64)$$

$$p_{1,\text{T},\text{refl}}^{\text{n,diff}} = p_{\text{refl},\text{T}}^{\text{diff}} \frac{2}{3} m \Gamma_{\nu-}^{\text{n,diff}} v_{\text{T}}, \quad (4.65)$$

$$p_{1,\text{T},\text{refl}}^{\text{n,maxw}} = p_{\text{refl},\text{T}}^{\text{maxw}} \frac{2}{3} m \Gamma_{\nu-}^{\text{n,maxw}} v_{\text{T}}, \quad (4.66)$$

$$p_{1,\text{inc}}^{\text{n,diff}} = \frac{m}{3} \frac{n_i K_{\text{cx},\text{m}}}{n_i K_{\text{cx},\text{m}} + n_e K_i} \left(n_n \left(\left(\frac{2T_i}{m} + u_\phi^2 + u_\tau^2 \right) M + M_{\nu\nu} \right) \right. \quad (4.67)$$

$$\left. - \frac{1}{n_i K_{\text{cx},\text{m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} \left(\left(\frac{4T_i}{m} u_\tau + u_\phi^2 u_\tau + u_\tau^3 \right) M + u_\tau M_{\nu\nu} \right) + \frac{\partial n_n}{\partial \nu} \left(\left(\frac{2T_i}{m} + u_\phi^2 + u_\tau^2 \right) M_\nu + M_{\nu\nu\nu} \right) \right) \right), \quad (4.68)$$

$$p_{1,\text{inc}}^{\text{n,maxw}} = \frac{m}{3} \left(n_n \left(\left(\frac{2T_n}{m} + u_{n\phi}^2 + u_{n\tau}^2 \right) M + M_{\nu\nu} \right) \right). \quad (4.69)$$

Term (2) pressure contributions

The contribution from Equation (3.64) is implemented in the code as

$$\begin{aligned} p_2 &= \frac{m}{3} n'_n u_{n\phi}^2 = \frac{m}{3} \frac{(n'_n u'_{n\phi})^2}{n'_n} \\ &= \frac{m}{3} \frac{\left((n'_n u'_{n\phi})_{\text{rec}} + (n'_n u'_{n\phi})_{\text{refl}}^{\text{diff/maxw}} + (n'_n u'_{n\phi})_{\text{T},\text{rec}} + (n'_n u'_{n\phi})_{\text{T},\text{refl}}^{\text{diff/maxw}} + (n'_n u'_{n\phi})_{\text{inc}}^{\text{diff/maxw}} \right)^2}{\left((n'_n)_{\text{rec}} + (n'_n)_{\text{refl}}^{\text{diff/maxw}} + (n'_n)_{\text{T},\text{rec}} + (n'_n)_{\text{T},\text{refl}}^{\text{diff/maxw}} + (n'_n)_{\text{inc}}^{\text{diff/maxw}} \right)}. \end{aligned} \quad (4.70)$$

Using the definition of RC_{n_n} (Eq. (4.18)) and $\text{RC}_{n_n}^{\frac{\partial n_n}{\partial \nu}}$ (Eq. (4.19)), the density contributions from Eq. (3.62) as written in Eq. (4.70) are given by

$$(n'_n)_{\text{rec}} = p_{\text{rec},\text{F}}^{\text{corr}} \text{RC}_{n_n} \frac{\Gamma_{\nu-}^i}{-M_\nu}, \quad (4.71)$$

$$(n'_n)_{\text{refl}}^{\text{diff}} = p_{\text{refl},\text{F}}^{\text{corr,diff}} \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n \text{RC}_{n_n} + \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \frac{\partial n_n}{\partial \nu} \text{RC}_{n_n}^{\frac{\partial n_n}{\partial \nu}} c_i \right), \quad (4.72)$$

$$(n'_n)_{\text{refl}}^{\text{maxw}} = p_{\text{refl},\text{F}}^{\text{corr,maxw}} n_n \text{RC}_{n_n}, \quad (4.73)$$

$$(n'_n)_{\text{T,rec}} = p_{\text{rec},\text{T}} \Gamma_{\nu-}^i \frac{2}{v_{\text{T}}}, \quad (4.74)$$

$$(n'_n)_{\text{T,refl}}^{\text{diff}} = p_{\text{refl},\text{T}} \Gamma_{\nu-}^{\text{n,diff}} \frac{2}{v_{\text{T}}}, \quad (4.75)$$

$$(n'_n)_{\text{T,refl}}^{\text{maxw}} = p_{\text{refl},\text{T}} \Gamma_{\nu-}^{\text{n,maxw}} \frac{2}{v_{\text{T}}}, \quad (4.76)$$

$$(n'_n)_{\text{inc}}^{\text{diff}} = \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n M - \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \left(\frac{\partial n_n}{\partial \tau} u_\tau M + \frac{\partial n_n}{\partial \nu} M_\nu \right) \right), \quad (4.77)$$

$$(n'_n)_{\text{inc}}^{\text{maxw}} = n_n M. \quad (4.78)$$

$$(4.79)$$

Using the definition of $\text{RC}_{u_{n,\phi}}$ (Eq. (4.21)) and $\text{RC}_{u_{n,\phi}}^{\frac{\partial n_n}{\partial \nu}}$ (Eq. (4.22)), the $(n'_n u'_{n\phi})$ terms from Eq. (3.67) as written in Eq. (4.70) are written as

$$(n'_n u'_{n\phi})_{\text{rec}} = p_{\text{rec},\text{F}}^{\text{corr}} \text{RC}_{u_{n,\phi}} c_i \frac{\Gamma_{\nu-}^i}{-M_\nu}, \quad (4.80)$$

$$(n'_n u'_{n\phi})_{\text{refl}}^{\text{diff}} = p_{\text{refl},\text{F}}^{\text{corr,diff}} \frac{n_i K_{\text{cx,m}}}{n_i K_{\text{cx,m}} + n_e K_i} \left(n_n \text{RC}_{u_{n,\phi}} c_i + \frac{1}{n_i K_{\text{cx,m}} + n_e K_i} \frac{\partial n_n}{\partial \nu} \text{RC}_{u_{n,\phi}}^{\frac{\partial n_n}{\partial \nu}} c_i^2 \right), \quad (4.81)$$

$$(n'_n u'_{n\phi})_{\text{refl}}^{\text{maxw}} = p_{\text{refl},\text{F}}^{\text{corr,maxw}} n_n \text{RC}_{u_{n,\phi}} c_n, \quad (4.82)$$

$$(n'_n u'_{n\phi})_{\text{T,rec}} = 0, \quad (4.83)$$

$$(n'_n u'_{n\phi})_{\text{T,refl}}^{\text{diff}} = 0, \quad (4.84)$$

$$(n'_n u'_{n\phi})_{\text{T,refl}}^{\text{maxw}} = 0, \quad (4.85)$$

$$(n'_n u'_{n\phi})_{\text{inc}}^{\text{diff}} = (n'_n)_{\text{inc}}^{\text{diff}} u_\phi, \quad (4.86)$$

$$(n'_n u'_{n\phi})_{\text{inc}}^{\text{maxw}} = (n'_n)_{\text{inc}}^{\text{maxw}} u_{n\phi}. \quad (4.87)$$

4.2.3 Net neutral energy flux density

Similar to the particle and parallel momentum flux, we write the net neutral energy flux density given by Eqs. (3.68) and (3.70) as follows:

$$Q^{\text{n,diff/maxw}} = Q_{\text{rec}}^{\text{n}} + Q_{\text{refl}}^{\text{n,diff/maxw}} + Q_{\text{T,rec}}^{\text{n}} + Q_{\text{T,refl}}^{\text{n,diff/maxw}} - Q_{\text{T,inc}}^{\text{n,diff/maxw}} \quad (4.88)$$

Using the definitions of RC_E (Eq. (4.24)) and $\text{RC}_E^{\frac{\partial n_n}{\partial \nu}}$ (Eq. (4.25)), the components are written as

$$Q_{\text{rec}}^{\text{n}} = \frac{1}{2} p_{\text{rec},\text{F}}^{\text{corr}} m \text{RC}_E c_{\text{i}}^3 \frac{\Gamma_{\nu-}^{\text{i}}}{-M_{\nu}}, \quad (4.89)$$

$$Q_{\text{refl}}^{\text{n,diff}} = \frac{1}{2} p_{\text{refl},\text{F}}^{\text{corr,diff}} m \frac{n_{\text{i}} K_{\text{cx},\text{m}}}{n_{\text{i}} K_{\text{cx},\text{m}} + n_{\text{e}} K_{\text{i}}} \left(-n_{\text{n}} \text{RC}_E c_{\text{i}}^3 + \frac{1}{n_{\text{i}} K_{\text{cx},\text{m}} + n_{\text{e}} K_{\text{i}}} \frac{\partial n_{\text{n}}}{\partial \nu} \text{RC}_E^{\frac{\partial n_{\text{n}}}{\partial \nu}} c_{\text{i}}^4 \right), \quad (4.90)$$

$$Q_{\text{refl}}^{\text{n,maxw}} = p_{\text{refl},\text{F}}^{\text{corr,maxw}} m n_{\text{n}} \text{RC}_E c_{\text{n}}^3, \quad (4.91)$$

$$Q_{\text{T,rec}}^{\text{n}} = p_{\text{rec},\text{T}} \Gamma_{\nu-}^{\text{i}} E_{\text{FC}}, \quad (4.92)$$

$$Q_{\text{T,refl}}^{\text{n,diff}} = p_{\text{refl},\text{T}} \Gamma_{\nu-}^{\text{n,diff}} E_{\text{FC}}, \quad (4.93)$$

$$Q_{\text{T,refl}}^{\text{n,maxw}} = p_{\text{refl},\text{T}} \Gamma_{\nu-}^{\text{n,maxw}} E_{\text{FC}}, \quad (4.94)$$

$$Q_{\text{inc}}^{\text{n,diff}} = \frac{n_{\text{i}} K_{\text{cx},\text{m}}}{n_{\text{i}} K_{\text{cx},\text{m}} + n_{\text{e}} K_{\text{i}}} \left(-n_{\text{n}} \left(\left(T_{\text{i}} + \frac{m}{2} (u_{\phi}^2 + u_{\tau}^2) \right) M_{\nu} + \frac{m}{2} M_{\nu\nu\nu} \right) + \right. \\ \left. \frac{1}{n_{\text{i}} K_{\text{cx},\text{m}} + n_{\text{e}} K_{\text{i}}} \left(\frac{\partial n_{\text{n}}}{\partial \tau} \left(\left(2T_{\text{i}} + \frac{m}{2} (u_{\phi}^2 + u_{\tau}^2) \right) u_{\tau} M_{\nu} + \frac{m}{2} u_{\tau} M_{\nu\nu\nu} \right) + \right. \right. \\ \left. \left. \frac{\partial n_{\text{n}}}{\partial \nu} \left(\left(T_{\text{i}} + \frac{m}{2} (u_{\phi}^2 + u_{\tau}^2) \right) M_{\nu\nu} + \frac{m}{2} M_{\nu\nu\nu\nu} \right) \right) \right), \quad (4.95)$$

$$Q_{\text{inc}}^{\text{n,maxw}} = -n_{\text{n}} \left(\left(T_{\text{n}} + \frac{m}{2} (u_{\text{n}\phi}^2 + u_{\text{n}\tau}^2) \right) M_{\nu} + \frac{m}{2} M_{\nu\nu\nu} \right). \quad (4.96)$$

4.3 Link between symbols and code variables

This section is specifically aimed at expert users of the AFN boundary conditions and future researchers that wish to improve them. It links the symbolic notation used in this document to the variable names in the code.

4.3.1 Pre-processor b2ab

Symbol	Code
m	<code>const.m</code>
$\mathbf{v}'$	<code>vi</code>
$\mathbf{v}$	<code>vo</code>
φ	<code>phiR</code>
φ'	<code>phi</code>
ϑ	<code>thetaR</code>
ϑ'	<code>thetai</code>
c_a	<code>cs</code>
$\delta_{\text{sh}}^{\text{pot}}$	<code>deltaP</code>
T_{e}	<code>te</code>
T_{i}	<code>ti</code>
v_{min}	<code>vmin</code>
ϑ_{max}	<code>thetamax</code>
$R_{\text{N}}(v', \vartheta')$	<code>TRIMRv</code>
$R_{\text{m}, }(v', \vartheta')$	<code>TRIMRmomparv</code>
$R_{\text{m},\perp}(v', \vartheta')$	<code>TRIMRmomperpv</code>
$R_{\text{p}}(v', \vartheta')$	<code>TRIMRpv1</code>
$R_{n_{\text{n}}}(v', \vartheta')$	<code>TRIMRpv2</code>
$R_{u_{\text{n}\phi}}(v', \vartheta')$	<code>TRIMRpv3</code>
$R_{\text{E}}(v', \vartheta')$	<code>TRIMREv</code>

4.3.2 Macroscopic reflection coefficients

Symbol	<u>in b2mod_create_afn_bc_tables</u>	<u>in b2mod_recycle</u>
RC_n	R1	rc1i
$RC_n^{\frac{\partial n_n}{\partial \nu}}$	R2	rc2i
$RC_{m,\phi}$	Rmx1	rmx1i
$RC_{m,\phi}^{\frac{\partial n_n}{\partial \nu}}$	Rmx2	rmx2i
$RC_{m,\tau}^{\frac{\partial n_n}{\partial \tau}}$	Rmy2	rmy2i
$RC_{m,\nu}$	Rmz1	rmz1i
$RC_{m,\nu}^{\frac{\partial n_n}{\partial \nu}}$	Rmz2	rmz2i
RC_E	RE1	re1i
$RC_E^{\frac{\partial n_n}{\partial \nu}}$	RE2	re2i
RC_p	Rp11	rp11i
$RC_p^{\frac{\partial n_n}{\partial \nu}}$	Rp12	rp12i
RC_{n_n}	Rp21	rp21i
$RC_{n_n}^{\frac{\partial n_n}{\partial \nu}}$	Rp22	rp22i
$RC_{u_n,\phi}$	Rp31	rp31i
$RC_{u_n,\phi}^{\frac{\partial n_n}{\partial \nu}}$	Rp32	rp32i

4.3.3 Runtime

General

<u>Symbol</u>	<u>Code</u>
e	qe
n_n	nnf / nn
b_θ	pit
b_ϕ	pit2
$p_{\text{rec},\text{F}}/p_{\text{refl},\text{F}}$	pf
$p_{\text{rec},\text{F}}^{\text{corr}}/p_{\text{refl},\text{F}}^{\text{corr}}$	pcorrf
$p_{\text{rec},\text{T}}/p_{\text{refl},\text{T}}$	pt
$R_{\text{fl}}^{\text{n}}/R_{\text{fl}}^{\text{i}}$	recyc0 / b2recyc
R_{m}	recycm
$\cos \alpha_\nu$	cosa
$\sin \alpha_\nu$	sina
ϕ	x
τ	y
ν	z
$u_{ }/u_{\text{n} }$	upf
$u_\phi/u_{\text{n}\phi}$	ux
$u_\tau/u_{\text{n}\tau}$	uy
$u_{\nu-}/u_{\text{n}\nu-}$	uz
$\mathcal{M}_\phi$	Mj
A_f	area
$n_i K_{\text{cx},\text{m}}$	vcx
$n_e K_i$	vion
$n_i K_{\text{cx},\text{m}} + n_e K_i$	vtot
v_{T}	vt
$\Gamma_H^{\text{n}} - Q^{\text{n}}$	q_conv
T_{n}	tf
T_i	tif
v_{T}	vt
E_{FC}	E_fc
Kn	Kn
L	L / L_macro_AFN
Kn^{bd}	Kn_b1
Kn^{bm}	Kn_b2
$c_{\text{n},th}$	c_th
c_i/c_{n}	cs
λ_{cx}	mfp
$\delta_{\text{sh}}^{\text{pot}}$	shpottauj

Moments of Maxwellian

<u>Symbol</u>	<u>Code</u>
M	i0l
M_ν	i1l
$M_{\nu\nu}$	i2l
$M_{\tau\nu}$	uy i1l
$M_{\phi\nu}$	ux i1l
$M_{\tau\tau\nu}$	(ti/mn + uy**2) i1l
$M_{\tau\nu\nu}$	uy i2l
$M_{\phi\tau\nu}$	ux uy i1l
$M_{\nu\nu\nu}$	i3l
$M_{\phi\nu\nu}$	ux i2l
$M_{\phi\phi\nu}$	(ti/mn + ux**2) i1l
$M_{\phi\phi\tau\nu}$	(ti/mn + ux**2) uy i1l
$M_{\tau\tau\tau\nu}$	(ti/mn + uy**2) uy i1l
$M_{\tau\nu\nu\nu}$	uy i3l
$M_{\phi\phi\nu\nu}$	(ti/mn + ux**2) i2l
$M_{\tau\tau\nu\nu}$	(ti/mn + uy**2) i2l
$M_{\nu\nu\nu\nu}$	i4l

CalcIncidentFluxesDiffusion/Maxwellian

<u>Symbol</u>	<u>Code</u>
$\Gamma_{m,\phi,inc}^{n,diff} / \Gamma_{m,\phi,inc}^{n,maxw}$	fmox
$\Gamma_{m,\tau,inc}^{n,diff} / \Gamma_{m,\tau,inc}^{n,maxw}$	fmoy
$\Gamma_{m,\nu-,inc}^{n,diff} / \Gamma_{m,\nu-,inc}^{n,maxw}$	fmoz
$\Gamma_{m, ,inc}^{n,diff} / \Gamma_{m, ,inc}^{n,maxw}$	fmomni
$Q_{inc}^{n,diff} / Q_{inc}^{n,maxw}$	feneni
$(n'_n)_{inc}^{diff} / (n'_n)_{inc}^{maxw}$	nni
$(n'_n u'_{n\phi})_{inc}^{diff}$	nnwwni
$p_{1,inc}^{n,diff} / p_{1,inc}^{n,maxw}$	pn

CalcRecycledFluxes

<u>Symbol</u>	<u>Code</u>
$\Gamma_{m,\phi,rec}^n$	fmox
$\Gamma_{m,\tau,rec}^n$	fmoy
$\Gamma_{m,\nu+,rec}^n + \Gamma_{m,\nu+,T,rec}^n$	fmoz
$p_{1,rec}^n + p_{1,T,rec}^n$	pnc
$(n'_n)_{rec} + (n'_n)_{T,rec}$	nnrec
$(n'_n u'_{n\phi})_{rec}$	nnwwnrec
$Q_{rec}^n + Q_{T,rec}^n$	fenec
$\Gamma_{m, ,rec}^n + \Gamma_{m, ,T,rec}^n$	fmomrec

CalcReflectedFluxesDiffusion/Maxwellian

Symbol	Code
$\frac{\Gamma_{m,\phi,\text{refl}}^{n,\text{diff}}}{\Gamma_{m,\phi,\text{refl}}^{n,\text{maxw}}}$	fmox
$\frac{\Gamma_{m,\tau,\text{refl}}^{n,\text{diff}}}{\Gamma_{m,\tau,\text{refl}}^{n,\text{maxw}}}$	fmoy
$\frac{\Gamma_{m,\nu+, \text{refl}}^{n,\text{diff}}}{\Gamma_{m,\nu+, \text{refl}}^{n,\text{maxw}}} + \frac{\Gamma_{m,\nu+, T, \text{refl}}^{n,\text{diff}}}{\Gamma_{m,\nu+, T, \text{refl}}^{n,\text{maxw}}}$	fmoz
$(n'_n)_{\text{refl}}^{\text{diff}/\text{maxw}} + (n'_n)_{T, \text{refl}}^{\text{diff}/\text{maxw}}$	nnrefl
$(n'_n u'_{n\phi})_{\text{refl}}^{\text{diff}/\text{maxw}}$	nnwwnrefl
$Q_{\text{refl}}^{n,\text{diff}/\text{maxw}} + Q_{T, \text{refl}}^{n,\text{diff}/\text{maxw}}$	fene
$\frac{\Gamma_{m, , \text{refl}}^{n,\text{diff}/\text{maxw}}}{\Gamma_{m, , \text{refl}}^{n,\text{diff}/\text{maxw}}} + \frac{\Gamma_{m, , T, \text{refl}}^{n,\text{diff}/\text{maxw}}}{\Gamma_{m, , T, \text{refl}}^{n,\text{diff}/\text{maxw}}}$	fmomrefl
$p_{1, \text{refl}}^{n,\text{diff}/\text{maxw}} + p_{1, T, \text{refl}}^{n,\text{diff}/\text{maxw}}$	pn

4.3.4 Other variables

In the final code implementation, certain terms are multiplied with additional factors that are not included in the equations as described here. These additional factors mostly relate to spatially hybrid options [25, 13, 1] and the SOLPS-ITER specific sign conventions:

- $I_s = \text{mpg}\%rcFc0r$ is a multiplier to indicate whether the outward normal coincides with the face normal (1.0) or not (-1.0).
- $R_m = \text{recycm}$ determines whether the thermal molecules are immediately dissociated at the target as fluid atoms (standard AFN), or launched kinetically as part of a spatially hybrid approach [25, 13].
- $R_{fl} = \text{fluid_frac_hyb}$ determines whether the recycled and reflected atoms come back as fluid ($\text{fluid_frac_hyb}=1$) or kinetic ($\text{fluid_frac_hyb}=0$). Used for spatially hybrid models. Values between 0 and 1 are allowed as well, to guarantee a smooth transition.
- $\text{sign}(u_\phi)$ makes sure that the momentum fluxes have the correct sign, regardless of the direction of the toroidal magnetic field.
- $A_f = \text{area}$ is the surface area of the boundary face in the finite volume mesh. It is used to transform the flux densities to the correct source strengths in the guard cells.

Chapter 5

User guide

5.1 Running with AFN boundary conditions in B2.5

5.1.1 Switches in b2mn.dat

The switches to control the behavior of the AFN model in B2.5 are listed here. The default values are given after the name of the switch.

`'b2stbr_recycle_afn'` (default = '0')

Main switch to enable the use of the AFN boundary conditions.

`'diffusion_bc_safeguard'` (default = '1')

When this switch is active, the code will automatically switch to using the Maxwellian approach instead of the diffusion approach, when the diffusion approach gives rise to negative incident fluxes.

`'remove_FC_el'` (default = '0')

When this switch is active, the 2-3 eV Franck-Condon energy given to the atoms originating from dissociating molecules is subtracted from the electrons. This switch should be active to guarantee global energy conservation. However, turning it on causes a strong localized electron heat sink, typically causing the electron temperature to drop to zero at the guard cells (see Ref. [24] Section 4.4). In all published AFN works, `'remove_FC_el' = 0`.

This is an important remaining shortcoming of the AFN model. An approximate fluid model for the molecules would solve this issue.

`'afn_bcs_use_coarse'` (default = '1')

When `'afn_bcs_use_coarse' = 0` the code will use AFN database such as `dw.fnn` with 100 discretization points for the toroidal Mach number, 134 points for the temperature, and 28 points for the sheath potential drop. When `'afn_bcs_use_coarse' = 1` the code will use AFN database such as `dw.fnn.coarse` with 11 discretization points for the toroidal Mach number, 134 points for the temperature, and 28 points for the sheath potential drop.

`'b2stbr_Kn_b1'` (default = '0.01')

When using `maxw(istra) = 2`, `b2stbr_Kn_b1` determines the Knudsen number below which the diffusion approximation is used.

`'b2stbr_Kn_b2'` (default = '0.1')

When using `maxw(istra) = 2`, `b2stbr_Kn_b2` determines the Knudsen number above which the Maxwellian approximation is used.

`'use_uy_uz_0'` (default = '0')

This switch controls the contribution from the tangential and normal bulk velocity of the ions or neutrals in the calculation of the incident fluxes. More research is needed on the role of these bulk velocities. It is a question of consistency, because these bulk velocities can be taken into account for the calculations of the incident fluxes, but the AFN databases are only tabulated for the toroidal Mach number.

`'use_uy_uz_1'` (default = '0')

This switch controls the contribution from the tangential and normal bulk velocity of the ions or neutrals in the calculation of the incident and recycled/reflected fluxes. More research is needed on the role of these bulk velocities. It is a question of consistency, because these bulk velocities can be taken into account for the calculations of the incident fluxes, but the AFN databases are only tabulated for the toroidal Mach number.

5.1.2 Inputs in `b2.neutrals.paramters`

`maxw(istra)` (default = 1)

When `'maxw(istra)' = 0`, the diffusion approximation for the incident velocity distribution of the neutrals is chosen. When `'maxw(istra)' = 1`, the Maxwellian approximation for the incident velocity distribution of the neutrals is chosen (recommended). When `'maxw(istra)' = 2`, the choice between diffusion or Maxwellian approximation is based on the local Knudsen number, depending on `b2stbr_Kn_b1` and `b2stbr_Kn_b2`.

`maxw_eff(istra)` (default = 1)

Relates to the option `CalcIncidentFluxesMinKn`, where the choice between diffusion and Maxwellian option is the same for each cell of the same stratum, but automatically selected based on the lowest Knudsen number along the boundary faces of that stratum. This option is depreciated and was never published. Not recommended.

`e_fc` (default = 3.0)

Franck-Condon energy (in eV) given to each atom resulting from a dissociating molecule.

`surf_mat(istra)` (default = 'C')

Surface material used to select the appropriate AFN database.

`accel_ion(istra)` (default = 0)

Determines whether the ions undergo sheath acceleration or not. For standard cases, `'accel_ion' = 1` for the targets and `'accel_ion' = 0` for PFR and main chamber wall boundaries.

`b2recyc(is,istra)` (default = 1.0)

Determines for each stratum and each species the fraction of incident particles that return to the domain (=1-pumped fraction). When `b2recyc` is not specified, `b2recyc = recyc`. The distinction between `b2recyc` and `recyc` is only relevant for spatially hybrid cases.

5.1.3 Default databases

Several AFN databases are available by default in `B2.5/Database/trim.data/refl.data/`. The original higher-resolution databases are `dc.fnn`, `dbe.fnn`, and `dw.fnn`. Next, all combinations of H, D, and T, with C, Be, Fe, Mo and W are available with a reduced resolution for the toroidal Mach number: `hc.fnn.coarse`, `hbe.fnn.coarse`, `hfe.fnn.coarse`, `hmo.fnn.coarse`, `hw.fnn.coarse`, `dc.fnn.coarse`, `dbe.fnn.coarse`,

dfc.fnn.coarse, dmo.fnn.coarse, dw.fnn.coarse, tc.fnn.coarse, tbe.fnn.coarse, tfe.fnn.coarse, tmo.fnn.coarse, and tw.fnn.coarse.

5.2 Creating new AFN databases with pre-processor b2ab

5.2.1 Running b2ab

Running `b2run (-s) b2ab` will create a new AFN database file based on the inputs in `b2ab.dat`, and place it in `b2ab.exe.dir`. The `b2ab.dat` file consists of a fixed-format header followed by optional arguments. The fixed-format header is:

```
*incident
'X'
*material
'Y'
```

where 'X' is the incident particle (currently H, D, or T) and 'Y' is the name of the wall material (C, Be, Cr, Li, W, ...). The code will then look for the TRIM file `X_on_Y`. It will first look in `modules/Eirene/Database/Surfacdata/TRIM/`. A user that does not have access to the EIRENE repository should request the specific TRIM file from an EIRENE developer (respecting the EIRENE license requirements), and place it in `modules/B2.5/Database/trim.data/refl.data.local/`.

The optional inputs are:

```
'b2abdr_ntp' (default = '1e5')
```

Number of Monte Carlo particles used to calculate the reflection coefficients that can not be evaluated analytically (see Section 3.2.6).

```
'b2abdr_nv' (default = '100')
```

Number of discretization points to evaluate the $\int_{v'=0}^{\infty} dv'$ the integrals from Section 4.1.

```
'b2abdr_ntheta' (default = '100')
```

Number of discretization points to evaluate the $\int_{\vartheta'=0}^{\pi/2} d\vartheta'$ the integrals from Section 4.1.

```
'b2abdr_nphi' (default = '180')
```

Number of discretization points to evaluate the $\int_{\varphi'=0}^{2\pi} d\varphi'$ the integrals from Section 4.1.

```
'b2abdr_nT' (default = '0')
```

Number of values for T_a in the AFN database.

```
'b2abdr_nM' (default = '0')
```

Number of values for $M_{a,\phi}$ in the AFN database.

```
'b2abdr_nSh' (default = '0')
```

Number of values for δ_{sh}^{pot} in the AFN database.

```
'b2abdr_Tmin' (default = '0.1')
```

Minimum value for T_a in the AFN database, expressed in eV.

```
'b2abdr_Tmax' (default = '1000')
```

Maximum value for T_a in the AFN database, expressed in eV.

`'b2abdr_Mmin' (default = '0.0')`

Minimum value for $M_{a,\phi}$ in the AFN database.

`'b2abdr_Mmax' (default = '1.1')`

Maximum value for $M_{a,\phi}$ in the AFN database.

`'b2abdr_Shmin' (default = '0.0')`

Minimum value for δ_{sh}^{pot} in the AFN database.

`'b2abdr_Shmax' (default = '6.0')`

Maximum value for δ_{sh}^{pot} in the AFN database.

`'b2abdr_Eomin' (default = '1e-6')`

Minimum incident particle energy (in eV) used throughout all calculations.

5.2.2 Allowed dimensions

In the current implementation, all databases used in a single run must have the same dimension. For example, it is possible to use `dw.fnn`, `dbe.fnn`, `tc.fnn`, and `tbe.fnn` in a single run. Likewise, `dw.fnn.coarse`, `dbe.fnn.coarse`, `tc.fnn.coarse`, and `tbe.fnn.coarse` can be used together. Mixing `tc.fnn.coarse`, and `tbe.fnn`, or another format with different dimensions, is not allowed. Therefore, it is recommended to stick to the standard formats when creating new databases.

5.2.3 Running with newly created databases

At runtime, the code will create target names based on the B2.5 atomic species, the `surf_mat` variables, and the `afn_bcs_use_coarse` switch. It will then first look for these names in `B2.5/Database/trim.data/refl.data.local/`, and then in `B2.5/Database/trim.data/refl.data/`. When a user has created a new database, for example D on Li, the file `dli.fnn.coarse` will be created in `b2ab.exe.dir`. The user then has to move this file to `B2.5/Database/trim.data/refl.data.local/`. Alternatively, new files that are expected to be needed by many users can be added to `B2.5/Database/trim.data/refl.data/` and pushed to the repository.

Chapter 6

Future work

The advanced fluid neutral boundary conditions should be kept up to date with any advancements in plasma boundary modeling, and further extensions for application to more realistic cases should be considered. Examples of possible extensions are:

- **Multi-species cases:** mixtures of H, D, or T have not yet been considered. Both for the transport model and the boundary conditions, the theory needs to be revisited for multi-species cases. For the boundary conditions, we suspect that little or no difficulty arises when using the truncated Maxwellian approximation for the incident atoms. However, when considering the diffusion approximation, the collisions between the different species must be taken into account.
- **Impurity species:** the same principles that were used here for the hydrogenic species could possibly lead to improved boundary conditions for the fluid modeling of impurity neutrals. It would then again be advised to use the truncated drifting Maxwellian assumption instead of the diffusion approximation for the incident atoms. For the recycling, separate contributions for each ionization stage must be considered, taking also into account the differences in sheath acceleration depending on the ionization stage.
- **Advanced sheath models:** we have used a simple sheath acceleration model, corresponding to what present-day EIRENE does. If more advanced sheath acceleration models become available in EIRENE, the recycling part of the AFN boundary conditions can be improved accordingly.
- **The role of perpendicular bulk velocities:** in the end, contributions from smaller perpendicular bulk velocities (both of the incident ions and incident atoms) were neglected to reduce the dimensionality of the pre-integrated databases. The effect of this simplification is not yet fully understood. However, good fluid-kinetic agreement was achieved in several AFN publications, including one where drifts were included in the plasma model [27]. Still, a more complete treatment of all velocity components could be desired in the future.
- **Neutral-neutral collisions:** when extending the fluid neutral models to include neutral-neutral collisions, the AFN boundary conditions must again be revisited. When using the truncated drifting Maxwellian assumption for the incident atoms, likely little or no change will be required. When using the diffusion approximation, the atom-atom collisions must be included in the derivation.
- **New reflection models:** if new reflection databases are released as a replacement for TRIM, the AFN boundary conditions should be extended to include those. Depending on the precise format of those new databases, most elements of the current procedure can be reused.
- **Fluid models for molecules:** if an approximate fluid description for the molecules is developed, the AFN boundary conditions can be easily adapted to provide realistic boundary conditions for these fluid

molecules. The fraction of incident ions and atoms that is thermally released as molecules is already calculated. What needs to be added is a model for the incident molecules that return as thermally released molecules.

When changing or extending the AFN boundary conditions, please also update this document. The LaTeX source files can be found in `solps-iter/doc/AFN_documentation/`.

Bibliography

- [1] M. Blommaert et al. “A spatially hybrid fluid-kinetic neutral model for SOLPS-ITER plasma edge simulations”. In: *Nuclear Materials and Energy* 19 (2019), pp. 28–33.
- [2] M. Blommaert et al. “Implementation of a consistent fluid-neutral model in SOLPS-ITER and benchmark with EIRENE”. In: *Contributions to Plasma Physics* 58.6-8 (2018), pp. 718–724.
- [3] Roland Chodura. “Plasma–wall transition in an oblique magnetic field”. In: *The Physics of Fluids* 25.9 (1982), pp. 1628–1633.
- [4] RH Cohen and DD Ryutov. “Sheath physics and boundary conditions for edge plasmas”. In: *Contributions to Plasma Physics* 44.1-3 (2004), pp. 111–125.
- [5] David Coulette and Giovanni Manfredi. “Kinetic simulations of the Chodura and Debye sheaths for magnetic fields with grazing incidence”. In: *Plasma Physics and Controlled Fusion* 58.2 (2016), p. 025008.
- [6] Wouter Dekeyser et al. *B2-B2. 5 code benchmarking*. Tech. rep. Forschungszentrum Juelich (Germany). Inst. fuer Energie-und Klimaforschung (IEK), 2011.
- [7] Gordon H Dunn. “Franck—Condon Factors for the Ionization of H₂ and D₂”. In: *The Journal of Chemical Physics* 44.7 (1966), pp. 2592–2594.
- [8] Gilberto Espinosa-Paredes et al. “Fractional neutron point kinetics equations for nuclear reactor dynamics”. In: *Annals of Nuclear Energy* 38.2-3 (2011), pp. 307–330.
- [9] John Greenwood. “The correct and incorrect generation of a cosine distribution of scattered particles for Monte-Carlo modelling of vacuum systems”. In: *Vacuum* 67.2 (2002), pp. 217–222.
- [10] D Heifetz et al. “A Monte-Carlo model of neutral-particle transport in diverted plasmas”. In: *Journal of Computational Physics* 46.2 (1982), pp. 309–327.
- [11] N. Horsten. “Fluid and Hybrid Fluid-Kinetic Models for the Neutral Particles in the Plasma Edge of Nuclear Fusion Devices”. PhD thesis. KU Leuven, 2019.
- [12] N. Horsten, G. Samaey, and M. Baelmans. “Development and assessment of 2D fluid neutral models that include atomic databases and a microscopic reflection model”. In: *Nuclear Fusion* 57.11 (2017), p. 116043.
- [13] N. Horsten et al. “Application of spatially hybrid fluid–kinetic neutral model on JET L-mode plasmas”. In: *Nuclear Materials and Energy* 27 (2021), p. 100969.
- [14] John R Lamarsh. “Introduction to nuclear reactor theory”. In: *(No Title)* (1966).
- [15] Ho Jin Park et al. “Generation of few-group diffusion theory constants by Monte Carlo code McCARD”. In: *Nuclear science and engineering* 172.1 (2012), pp. 66–77.
- [16] Pamela Piovezan et al. “Heavy steel reflector evaluation using diffusion theory”. In: (2013).
- [17] D Reiter. *The EIRENE code user manual version: 11/2009*. 2009.
- [18] ME Rensink et al. “A comparison of neutral gas models for divertor plasmas”. In: *Contributions to Plasma Physics* 38.1-2 (1998), pp. 325–330.

- [19] J Riemann et al. “Navier-Stokes Neutral and Plasma Fluid Modelling in 3D”. In: *Contributions to Plasma Physics* 44.1-3 (2004), pp. 35–38.
- [20] PC Stangeby. “Plasma sheath transmission factors for tokamak edge plasmas”. In: *The Physics of fluids* 27.3 (1984), pp. 682–690.
- [21] PC Stangeby. “The Chodura sheath for angles of a few degrees between the magnetic field and the surface of divertor targets and limiters”. In: *Nuclear Fusion* 52.8 (2012), p. 083012.
- [22] DD Tskhakaya and L Kos. “Comprehensive kinetic analysis of the plasma-wall transition layer in a strongly tilted magnetic field”. In: *Physics of Plasmas* 21.10 (2014).
- [23] DD Tskhakaya et al. “Theory of the plasma sheath in a magnetic field parallel to the wall”. In: *Physics of plasmas* 12.10 (2005).
- [24] W. Van Uytven. “Development of Fluid and Hybrid Fluid-Kinetic Atom Models for Plasma Boundary Simulations and Assessment in Reactor-Relevant Conditions”. PhD thesis. KU Leuven, 2023.
- [25] W. Van Uytven et al. “Advanced spatially hybrid fluid-kinetic modelling of plasma-edge neutrals and application to ITER case using SOLPS-ITER”. In: *Contributions to Plasma Physics* (2022), e202100191.
- [26] W. Van Uytven et al. “Assessment of advanced fluid neutral models for the neutral atoms in the plasma edge and application in ITER geometry”. In: *Nuclear Fusion* (2022).
- [27] Wim Van Uytven et al. “Effect of drifts and currents on the validity of a fluid model for the atoms in the plasma edge”. In: *Nuclear Materials and Energy* 33 (2022), p. 101255.